Effects of Subsidies on Welfare and Market Structure in the U.S. Broadband Industry*

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Abstract

I study how demand-side and supply-side broadband subsidies affect welfare, prices, and market structure. Event studies on a state-year panel of 52 U.S. states show that the Affordable Connectivity Program (ACP) generated significant price reductions—particularly for high-speed plans—and a compositional shift toward higher-speed offerings. The Broadband Equity, Access, and Deployment (BEAD) program produced smaller but persistent price reductions and concentration declines with a two-to-three year lag, consistent with infrastructure deployment timelines. I estimate a structural model of differentiated broadband products with endogenous provider entry on 220,814 products across 69,085 census tracts. A 75 percent BEAD fixed-cost subsidy induces up to 341,000 new provider-tract entries and raises consumer surplus by \$0.7 billion per month; a \$30 ACP discount raises consumer surplus by \$3.1–\$3.5 billion monthly at substantially higher recurring fiscal cost. BEAD generates strictly positive net present value; ACP operates primarily as an income transfer. The two programs are economic complements.

Keywords: Broadband; public subsidies; market structure; welfare analysis; digital divide; structural econometrics.

JEL classification: D22, L13, L51, L52, L96, L98.

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1 Introduction

Broadband access has become a prerequisite for economic participation, yet the digital divide remains wide. In 2020, roughly 21 million American households lacked fixed broadband service, a gap that falls disproportionately on low-income and rural communities. Congress responded with the Infrastructure Investment and Jobs Act (IIJA) of 2021, which allocated approximately $65 billion to broadband and created two flagship programs: the Affordable Connectivity Program (ACP), a $14.2 billion demand-side subsidy providing eligible low-income households with monthly discounts of up to $30 on broadband bills, and the Broadband Equity, Access, and Deployment (BEAD) program, a $42.5 billion supply-side grant funding deployment in unserved and underserved areas. These programs share a goal (wider, more affordable broadband) but operate through fundamentally different economic mechanisms, raising a natural question: how much does each program improve welfare, for whom, and at what fiscal cost?

The economic distinction between the two programs is sharper than it first appears. A demand-side subsidy like ACP lowers the effective price for eligible households, expanding take-up among price-sensitive consumers. But it also alters competitive dynamics: ISPs competing for newly subsidized subscribers may respond by lowering list prices, expanding higher-speed offerings, or both. These market-equilibrium responses are invisible to program evaluations that treat ACP purely as a household transfer. A supply-side subsidy like BEAD reduces the fixed cost of deploying infrastructure, inducing entry in markets where providers previously found deployment unprofitable. That entry, in turn, disciplines incumbent pricing over time. The two channels interact: ACP-driven demand growth raises variable profits and can encourage entry at the margin, while BEAD-induced entry reduces market power and improves affordability. Yet their timing, magnitudes, and fiscal costs differ substantially. Disentangling them requires both credible reduced-form identification and a structural model that allows for endogenous entry and product positioning.

I address this gap by combining event study evidence from a state-year panel with a structural IO model of the U.S. broadband market. In the reduced-form analysis, I exploit a panel of 52 U.S. states observed annually from 2014 to 2025, using continuous treatment intensities (state-level ACP eligibility exposure and BEAD allocation per unserved location) in two-way fixed-effects event studies. In the structural analysis, I construct a new local product-level dataset from FCC Form 477 deployment records, FCC Urban Rate Survey prices, and American Community Survey demographics, covering 220,814 broadband products in 69,085 census tracts as of December 2020. I estimate a nested logit demand system following Berry [1994] and Berry et al. [1995], recover Bertrand-Nash markups and marginal costs as in Nevo [2001], and estimate fixed costs of entry using the scalable discrete-game bounds of Fan and Yang [2024], which avoids solving the full entry game during estimation. The estimated model is used to simulate counterfactual policies and conduct cost-benefit analysis.

Three main findings emerge. First, the event study evidence shows that ACP generated significant and economically large market effects beyond a simple income transfer. In states at the 75th percentile of ACP eligibility exposure relative to those at the 25th percentile, broadband prices fell by roughly 30–45 log points for average and high-speed plans at peak response. Plan composition shifted substantially toward higher-speed offerings, and provider concentration declined. These patterns cannot be explained by a pure demand-transfer model. They suggest that ACP triggered competitive repricing and product repositioning by ISPs: subsidizing price-sensitive consumers expanded the contestable market and intensified competition for that customer base. BEAD produced different, smaller effects that materialize with a two-to-three year lag, consistent with the capital-intensive nature of infrastructure deployment.

Second, structural counterfactuals quantify the equilibrium mechanisms behind these patterns. A 75 percent BEAD fixed-cost subsidy induces between 115,000 and 341,000 new provider-tract entries, raising consumer surplus from a baseline of \$2.18–\$2.65 billion per month to \$2.86–\$3.36 billion. A \$30 ACP discount raises consumer surplus to \$5.31–\$5.91 billion per month through price reductions that expand take-up. The two policies are complements: ACP improves affordability where infrastructure exists; BEAD creates infrastructure where it does not. Because the structural model holds gross prices fixed, it captures the demand-expansion channel but not the competitive repricing channel identified in the event study. The structural ACP welfare gains are therefore conservative.

Third, cost-benefit analysis reveals a fundamental difference in fiscal efficiency. The 75 percent BEAD fixed-cost reduction has an NPV of \$59–\$223 billion at a 3 percent discount rate over 25 years and a benefit-cost ratio of 3.5–12.7. ACP's BCR of 0.63–1.37 for the \$30 discount reflects its nature as a recurring transfer program: the fiscal outlay is permanent, and a meaningful share goes to households who would have subscribed regardless of the marginal subsidy. The structural BCR is a lower bound on ACP's true return, since competitive repricing spillovers are not captured. Even so, the gap between BEAD's and ACP's benefit-cost ratios is large enough to survive reasonable corrections.

The rest of the paper proceeds as follows. Section 2 presents the contributions and situates the paper in related literatures. Section 3 describes the institutional setting, data, and local market structure. Section 4 presents the reduced-form evidence. Section 5 develops the structural model, including timing assumptions, identification, and a discussion of multiple equilibria; Appendix F discusses measurement assumptions in detail. Section 6 reports estimation results and structural parameter diagnostics. Section 7 presents counterfactual policy simulations and distributional effects across market types. Section 8 reports the cost-benefit analysis, including a sensitivity analysis with respect to discount rate and infrastructure horizon. Section 9 concludes.

2 Contributions and Related Literature

This paper connects four strands of literature and makes three distinct contributions. I present each contribution alongside the literature it builds on and departs from.

Reduced-form identification of market equilibrium responses to broadband subsidies. Using a continuous-treatment event study on a 52-state panel spanning 2014–2025, I document that ACP generated significant competitive responses in the broadband market: price reductions, product upgrading, and concentration declines that go well beyond what a pure income-transfer evaluation would detect. The identification relies on pre-existing cross-state variation in ACP eligibility exposure and BEAD funding intensity, with state and year fixed effects and state-specific trends for ACP. The pre-period placebo tests are broadly supportive of the parallel-trends assumption.

The reduced-form identification strategy builds on the staggered difference-in-differences literature [Sun and Abraham, 2021, Callaway and Sant'Anna, 2021] and the Bartik-style continuous treatment approach of Goldsmith-Pinkham et al. [2020]. The event study complements a growing literature documenting market failures in U.S. broadband. High prices, limited provider choice, and geographic concentration are well-established features of local broadband markets [Flamm and Varas, 2022]. Greenstein and Mazzeo [2006] show that differentiation strategies among local ISPs shape competition and pricing in last-mile markets, while Xiao and Orazem [2011] find that entry by a fourth provider has minimal competitive effects, suggesting the market reaches effective saturation at low provider counts. Espin and Rojas [2024] and Kearns [2024] study how competition and infrastructure investment interact with digital divide outcomes. Lin, Tang, and Xiao [2024] address endogeneity in cellphone deployment games, documenting the importance of strategic interactions in telecommunications entry. The digital divide literature emphasizes that broadband adoption gaps across income, geography, and race have significant downstream consequences for labor markets, education, and health [Goolsbee and Klenow, 2002, Forman et al., 2005, Goldfarb and Prince, 2008]. Unlike existing broadband event studies, I directly identify supply-side competitive responses—price reductions and product repositioning—triggered by a demand-side subsidy, establishing the general-equilibrium spillover mechanism that the structural model is designed to quantify.

Structural evaluation of demand-side and supply-side broadband subsidies with endogenous entry. I estimate a nested logit demand system at the census-tract level using Berry inversion and BLP instruments, recover Bertrand-Nash markups and marginal costs, and bound provider fixed costs using the scalable discrete-game method of Fan and Yang [2024]. The structural model is essential for three tasks the reduced-form approach cannot accomplish: (a) welfare decomposition into consumer surplus, producer surplus, and fiscal cost; (b) counterfactual evaluation of alternative policy designs not observed in the data; and (c) quantification of long-run equilibrium entry responses and their feedback into prices and product availability. The model is estimated on a new tract-level dataset covering 220,814 broadband products across 69,085 U.S. census tracts.

The demand estimation builds on the seminal work of Berry [1994] and Berry, Levinsohn, and Pakes [1995], who develop the BLP framework for estimating discrete-choice demand systems with endogenous prices. Nevo [2001] provides a clean application to the cereal industry and demonstrates markup recovery from the Bertrand-Nash first-order conditions. Conlon and Gortmaker [2020] offer modern computational implementations. Crawford, Shcherbakov, and Shum [2019] develop the latent-market-share construction used in this paper to infer product-level shares from aggregate adoption data in cable television markets. Berry and Haile [2014] establish nonparametric identification in differentiated-product markets using market level data. Petrin [2002] and Gentzkow [2007] provide additional applications demonstrating the power of structural demand estimation for welfare analysis. On the demand side, Rosston, Savage, and Waldman [2010] estimate household demand for broadband using a discrete choice framework, finding price elasticities in the range of -1 to -2 at the household level, while Nevo, Turner, and Williams [2016] document substantial consumer surplus from high-speed plans.

The supply-side framework draws on the discrete-game entry literature. Bresnahan and Reiss [1991] pioneered the empirical study of entry in oligopolistic markets. Berry [1992] estimates an ordered entry model for airlines. Mazzeo [2002] studies product-type selection in motel markets with heterogeneous competitors. Seim [2006] estimates a model of geographic entry with endogenous product-type choices in video rental markets. Fan [2013] uses a structural model of product-line and pricing decisions to study newspaper ownership consolidation. Ciliberto and Tamer [2009] develop partial-identification methods for entry games with multiple equilibria, and Tamer [2003] establishes identification in incomplete discrete games. Fan and Yang [2024] extend this framework to settings with many firms and many discrete decisions, developing moment-inequality bounds that avoid solving the full game during estimation—a scalability that is essential for broadband markets, where the typical census tract has over ten feasible provider-tier positions.

Cost-benefit analysis comparing supply-side and demand-side broadband policy. I show that BEAD and ACP are not interchangeable and serve fundamentally different roles. BEAD's one-time capital investment generates durable infrastructure benefits with strongly positive net present value. ACP's recurring transfer provides immediate affordability relief but requires permanent fiscal commitment. The analysis reveals that demand-side subsidies in oligopolistic markets may generate competitive spillovers—price reductions and product upgrading—that improve their social return beyond what simple transfer-program evaluations detect.

This CBA connects to the literature on subsidy incidence in imperfectly competitive markets. The canonical competitive-market analysis of Gruber [1994] and the broader literature on transfers in kind [Currie and Gahvari, 2008] establish that incidence depends on relative supply and demand elasticities. In oligopolistic markets, however, strategic firm responses can substantially alter this logic. When a demand-side subsidy expands the contestable market and intensifies competition, incumbent firms may reduce prices and upgrade products beyond what the direct discount effect predicts, improving surplus for unsubsidized consumers as well. This paper documents this mechanism empirically and quantifies it structurally. The connection between demand-side subsidies and competition connects to the broader literature on competition and productivity [Aghion et al., 2005], and has direct implications for how federal broadband interventions should be evaluated.

3 Industry Background and Data

3.1 Industry Background

3.1.1 Current Landscape

The U.S. broadband industry has grown substantially over the past two decades and now touches nearly every dimension of household economic life. This paper focuses on residential broadband, which has attracted sustained federal attention and large public investments. Broadband is offered in several forms, with internet service providers (ISPs) competing through differences in speed, data limits, and pricing.

Traditionally, local telephone and cable companies have been the primary providers of broadband services. Telephone companies deliver broadband through Digital Subscriber Line (DSL) technology, utilizing older copper wires installed on telephone poles. Cable companies provide broadband over coaxial cables originally deployed for television. Today, the industry has expanded with more firms, including large nationwide providers and smaller local operators. Local governments participate in the market in most states, except in a few states where legislation restricts municipal provision.[1] Most U.S. households connect through one of the following technologies: DSL, cable, fixed wireless, or fiber optics.[2] Fixed wireless transmits the internet via radio signals from a fixed tower to a receiver at the customer's location, making it practical for rural areas where wired infrastructure is costly. Fiber-optic broadband uses thin glass or plastic strands to transmit data as light pulses, offering superior speed, reliability, and bandwidth. It supports symmetrical upload and download speeds, low latency, and high-demand applications such as streaming, cloud services, and remote work. Fiber deployment, however, requires substantial investment and is primarily available in urban and suburban areas.

Research on U.S. broadband consistently returns two findings: service is expensive relative to household income, and in most local markets consumers face limited provider choice [Flamm and Varas, 2022]. In my December 2020 sample, the average census tract has 2.36 providers offering fixed broadband, and the median provider HHI—computed within tracts using provider market shares—is 5,712, well above the 2,500 threshold traditionally associated with concentrated markets. Fiber-optic coverage, which offers the best performance and reliability, is concentrated in urban and suburban areas; rural tracts disproportionately rely on DSL over aging copper infrastructure, which has lower speeds and reliability.

These market structure conditions underlie what the federal government calls the digital divide: the gap between households with reliable internet access and those without. Income, geography, and digital literacy all predict which side of that gap a household falls on, and the consequences extend to job search, healthcare, and schooling [Forman et al., 2005, Goldfarb and Prince, 2008, Goolsbee and Klenow, 2002]. The federal government has responded with a range of subsidy programs: some target affordability directly; others subsidize infrastructure deployment in areas where private investment has not materialized.

3.1.2 Federal Broadband Subsidies

Congress passed the Infrastructure Investment and Jobs Act (IIJA) in 2021, allocating roughly $65 billion to broadband. The two main programs are the Affordable Connectivity Program (ACP), which subsidized household internet bills directly, and the Broadband Equity, Access,

[1] For instance, Texas state law significantly restricts municipal governments from directly providing broadband services to residents (see Baller, Stokes and Lide, 2020; BroadbandNow, 2023). Wilson [2025] finds that municipal entry crowds out private ISP investment, suggesting that public provision involves a market-structure trade-off beyond the deployment-gap rationale that motivates programs like BEAD.

[2] See Appendix for illustrative images.

and Deployment (BEAD) program, which funds deployment in underserved areas. Of the total, \$14.2 billion went to ACP and \$42.45 billion to BEAD.

ACP was introduced as a successor to the Emergency Broadband Benefit (EBB). The EBB program, which ran from February to late 2021, reached nearly nine million low-income households with a subsidy of up to \$50 per month. The ACP enrolled 23 million households at its peak, offering a monthly discount of up to \$30 (\$75 on qualifying Tribal lands). Eligible households could also receive a one-time discount of up to \$100 to purchase a connected device.[3] ACP funding was exhausted in June 2024 when appropriations were not renewed by Congress. The program's termination is particularly relevant for this paper: the period 2022–2024 represents ACP's full operational lifecycle, from launch through exhaustion, and is fully covered by the reduced-form event study panel. ACP's termination also raises the immediate policy question of whether a successor program is warranted and, if so, what form it should take. The welfare and fiscal-efficiency analysis in this paper directly informs that debate.

The BEAD program represents the largest federal investment targeting broadband providers directly. It allocates funds to states based on the number of underserved locations in each jurisdiction. The FCC defines underserved locations as broadband-serviceable locations that either lack broadband service entirely or lack access to reliable service at speeds of at least 25/3 Mbps with latency below 100 milliseconds.[4] Allocations are determined by federal formulas, so states cannot independently influence the amount of funding they receive. As of 2025, states were in various stages of BEAD implementation, with initial subgrantee selection underway and deployment expected to extend through 2026 and beyond. The structural model uses a pre-IIJA baseline and provides counterfactual predictions that are best understood as the program's steady-state equilibrium effects once deployment is complete, rather than as predictions of specific intermediate-year outcomes.

3.1.3 Economic Rationale

The economic rationale for both programs rests on market failures in residential broadband. On the demand side, low income and limited digital literacy create participation barriers: even where broadband is available, many households cannot afford monthly plans. The ACS reports that in 2020, roughly 15 percent of U.S. households with broadband access reported being unable to afford their current plan, and broadband subscription rates among households below 200 percent of the federal poverty line were roughly 20 percentage points below the national average. A demand-side subsidy addresses this access gap directly, lowering the

[3] https://www.fcc.gov/acp
[4] https://broadbandusa.ntia.doc.gov/

effective price to a level where price-constrained households are willing to subscribe.

On the supply side, the market failure is different: fixed deployment costs are high enough to make broadband deployment privately unprofitable in low-density or low-income areas, even when the social benefits of connectivity are large. The tension between private and social returns to infrastructure investment is a classic rationale for public subsidy [Bresnahan and Reiss, 1991, Kearns, 2024]. An important caveat is that public supply-side intervention may also crowd out private entry: Wilson [2025] finds that municipal broadband provision reduces private ISP investment in affected markets, a general-equilibrium cost that standard program evaluations omit. BEAD directly addresses the deployment margin by reducing the provider's private cost of entry, but the net welfare effect depends on how much private investment would have occurred in its absence.

The economic distinction matters for program design and evaluation. A demand-side subsidy like ACP has no direct effect on whether infrastructure is available in the first place; it can only stimulate adoption in markets where ISPs already offer service. In unserved areas—where infrastructure does not exist—no level of demand-side subsidy can induce adoption. BEAD, by contrast, targets the deployment decision directly. But BEAD cannot improve affordability in markets where infrastructure already exists and prices are the binding constraint. The two programs are thus economic complements: each addresses a market failure the other cannot.

An additional consideration is that the broadband market is oligopolistic, not perfectly competitive. Most U.S. households have access to at most two or three fixed-broadband providers, and many face a single dominant ISP [Flamm and Varas, 2022]. In this setting, demand-side subsidies may have competitive spillovers: a subsidy that expands the addressable market can intensify competition for newly subsidized customers, inducing price reductions and product improvements that benefit all consumers, not just program recipients. This general-equilibrium channel is theoretically important but rarely quantified in program evaluations. Documenting and measuring it is one of the central contributions of this paper.

3.2 Data Sources and Estimation Samples

The empirical analysis uses two complementary samples. The first is a state-year policy panel used to estimate reduced-form effects of federal broadband funding exposure. The second is a newly constructed local product-level sample used to estimate demand, recover marginal costs, and discipline the structural counterfactuals. The state-year panel combines IIJA and BEAD allocation records with annual broadband prices, product characteristics, and state demographics. The local sample combines FCC Urban Rate Survey prices, FCC Form 477 deployment records, FCC broadband adoption data, and ACS demographics.

For the local demand and supply estimation, a market is a census tract and a product is a canonical provider[5] × speed-tier pair. The baseline sample is constructed for December 2020, before the implementation of the major IIJA broadband programs. Speed tiers follow the IIJA broadband standard: the high-speed tier contains products with download speed of at least 25 Mbps and upload speed of at least 3 Mbps; the low-speed tier contains all other fixed broadband products. Appendix A reports file-level details, merge keys, sample restrictions, and data-source links.

3.3 URS Price Matching

Monthly prices are assigned to each observed and potential broadband product using plan-level observations from the FCC Urban Rate Survey (URS). Products are defined at the census-tract–canonical-provider–speed-tier level. URS prices are converted to constant 2025 dollars using CPI adjustment factors.

For each product, the assignment procedure searches for comparable URS plans using a six-level brand-first hierarchy. The algorithm first searches for plans in the same state, offered by the same canonical provider, and in the same speed tier. If no such plan is available, the search expands sequentially to national provider-tier matches, provider-specific matches that ignore the speed tier, state-level tier matches pooling across providers, and finally national tier-level matches across all providers.

Within the first non-empty match category, the assigned price is the URS `Total Charge` of the plan whose advertised download speed is closest to the FCC product's average advertised download speed. The procedure additionally records the matched URS plan speed, the percentage distance between FCC and URS plan speeds, the match level used, and an indicator for successful price imputation.

The same assignment rule is applied to both observed and potential products. For observed products, average advertised download speed is computed from Form 477 block-level deployment records aggregated to the tract–provider–speed-tier level. For potential products, characteristics are assigned using the provider's state-level average within the corresponding speed tier. Products without a valid URS match are retained with missing prices and flagged as unmatched.

[5] Raw FCC provider names contain subsidiary, regional, and legacy operating names belonging to the same corporate family. I consolidate these into a normalized corporate-family identifier before constructing products, assigning prices, calculating market shares, and analyzing entry. For example, Southwestern Bell, Indiana Bell, and BellSouth are mapped to `att`; Time Warner Cable, Bright House, and Spectrum to `charter`; CenturyLink, Quantum Fiber, and Lumen to `centurylink`; and Starlink and SpaceX to `starlink`.

The hierarchy achieves a high degree of specificity in practice. For the December 2020 sample, 40.8% of products are matched at the most granular level (state × brand × tier), while 75.5% are matched using state-specific information. Only 1.4% of products rely on the broadest national-tier fallback. Overall, the procedure achieves a 100% price imputation rate for the 673,110 product observations in the estimation sample. Additional implementation details are provided in Appendix C.

3.4 Market Definition and Sample Construction

A market is a U.S. census tract. This definition is both fine enough to capture genuine geographic heterogeneity in competition and coarse enough to be measurable with publicly available data. Census tracts contain on average 4,000 residents and are designed by the Census Bureau to have roughly homogeneous socioeconomic characteristics. Broadband markets have natural local boundaries: ISPs compete for households within a given geography, and the cost of expanding infrastructure is increasing in distance. Tract-level markets have become the standard unit of analysis in empirical work on local broadband competition [Flamm and Varas, 2022, Kearns, 2024].

A product is a canonical-provider–speed-tier pair within a tract. I consolidate provider names to canonical corporate families (e.g., all AT&T subsidiaries to `att`) and define speed tiers using the IIJA broadband standard: the high-speed tier ($g = H$) contains products with download speed of at least 25 Mbps and upload speed of at least 3 Mbps; the low-speed tier ($g = L$) contains all other fixed broadband offerings. This two-tier classification captures the most policy-relevant distinction—products meeting versus not meeting the federal minimum broadband standard—while remaining tractable for the entry game.

Market size M_t is the number of households in the tract from the 2020 American Community Survey. The inside market (broadband subscribers) and outside option (non-subscribers) are defined relative to this population.

3.5 Constructing Product-Level Market Shares

The central measurement challenge in estimating broadband demand is the absence of publicly available product-level market shares. The FCC reports where providers offer service, the technologies they use, and advertised speeds, but does not publicly report subscribers by provider-speed product in each census tract. This missing object is essential for demand estimation.

I construct product-level shares by combining FCC adoption information with local product availability and characteristics, following the logic of Crawford et al. [2019]. The

FCC reports tract-level fixed broadband adoption by speed threshold in binned ranges. For example, a tract in the $400 < x \leq 600$ bin has an adoption rate between 40 and 60 percent for the relevant fixed broadband speed category.

I use these binned adoption measures to recover aggregate adoption by speed tier. Let S_t^H denote the high-speed adoption share in tract t, and S_t^L the low-speed adoption share. In the baseline construction, I assign S_t^H using the midpoint of the FCC adoption interval, and recover low-speed adoption as the residual:

$$S_t^L = S_t^{\text{fixed}} - S_t^H, \tag{3.1}$$

where S_t^{fixed} is total fixed-broadband adoption in tract t. The outside-option share is

$$s_{0t} = 1 - S_t^H - S_t^L. \tag{3.2}$$

The next step allocates tier-level adoption across providers available in the tract. I first estimate an aggregate adoption equation at the tract level:

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t'\kappa + D_t'\lambda + u_t, \tag{3.3}$$

where $\bar{X}_t$ contains tract-level averages of product characteristics and D_t contains ACS demographic controls. Within each tract and speed tier, I allocate the FCC-implied adoption share across available products using logit shares implied by predicted utilities:

$$\hat{s}_{j|g,t} = \frac{\exp(\hat{\delta}_{jgt})}{\sum_{\ell \in \mathcal{J}_{gt}} \exp(\hat{\delta}_{\ell gt})}, \tag{3.4}$$

where $\mathcal{J}_{gt}$ is the set of products in tier g available in tract t. The product-level market share is then

$$\hat{s}_{jgt} = \hat{s}_{j|g,t} S_t^g. \tag{3.5}$$

By construction, product shares sum to total fixed-broadband adoption in each tract. The constructed shares should be interpreted as inferred product shares, not directly observed subscriber counts. Appendix D describes the treatment of FCC adoption bins, the variables used in (3.3), and diagnostics for the share-construction procedure.

Table 1: Summary Statistics: Local Demand and Supply Sample

Variable	Mean	Std. Dev.	P25	P75
Product-level variables				
Price ($/month)	71.65	25.34	49.99	86.17
Download speed (Gbps)	0.337	0.713	0.025	0.300
Upload speed (Gbps)	0.065	0.136	0.003	0.030
High-speed indicator	0.624	–	–	–
Product market share $\hat{s}_{jt}$	0.259	0.152	0.159	0.324
Within-tier share $\hat{s}_{j\|g,t}$	0.313	0.166	0.202	0.383
Tract-level variables				
Outside-option share s_{0t}	0.171	0.134	0.100	0.300
Products per tract	3.20	–	2	4
Providers per tract	2.36	–	2	3
Provider HHI	5,712	2,584	3,694	6,394
Households	1,679	820	1,126	2,089
Population	4,511	2,325	2,967	5,601
Median household income ($)	67,152	33,805	44,318	82,159
Renter share	0.364	0.228	0.183	0.509
College share	0.311	0.193	0.161	0.425
Age 65+ share	0.164	0.080	0.114	0.200
Broadband adoption share	0.668	0.171	0.559	0.803

Notes: The sample contains 220,814 tract-provider-tier products in 69,085 census tracts and 1,732 canonical providers. The high-speed tier is defined as download speed of at least 25 Mbps and upload speed of at least 3 Mbps. Product shares are inferred using FCC adoption information and product-level availability (Section 3.5). Provider HHI is computed within tracts using provider shares among fixed-broadband subscribers and is scaled from 0 to 10,000.

3.6 Descriptive Statistics

3.7 Local Market Structure

The market structure of the local broadband industry shapes both the identification strategy and the interpretation of the structural results. Table 2 summarizes the distribution of provider counts in the estimation sample.

Nearly a quarter of census tracts in the sample have a single fixed-broadband provider, and almost two-thirds have at most two. This concentration is the fundamental market failure the federal programs address. Monopoly tracts account for 21.96 percent of households—a substantial fraction facing no intra-market price competition for fixed broadband. The moderate share of tracts with three or more providers (36.73 percent) reflects that competitive broadband markets, while they exist, are not the norm. Rural tracts and tracts in low-income counties are disproportionately in the monopoly or duopoly categories.

Table 2: Provider Competition in the Local Sample (December 2020)

Number of providers in tract	Share of tracts	Share of households
One provider	23.10%	21.96%
Two providers	40.16%	39.22%
Three or more providers	36.73%	38.82%

Notes: Provider counts based on canonical providers with positive constructed product shares in the tract. Sample: 69,085 census tracts in December 2020.

Table 3 documents the structure of national supply. The top nine providers jointly account for 74.3 percent of implied subscribers in the sample, with AT&T and Comcast alone holding roughly 46 percent. This concentration at the national level contrasts with fragmented local markets: while a small number of large providers dominate nationally, their geographic footprints do not overlap sufficiently to create competitive local markets. This geographic segmentation is precisely why supply-side deployment subsidies like BEAD—targeted at specific unserved locations rather than national incumbents—are better suited to the entry margin than demand-side programs.

Table 3: National Market Shares by Provider (December 2020)

Provider	Implied subscribers (M)	Share of inside good (%)	Share of households (%)	Tracts
AT&T	27.49	28.21	23.69	33,344
Comcast	17.28	17.74	14.90	28,950
Verizon	10.53	10.80	9.07	12,070
Frontier	7.83	8.03	6.75	9,549
Cox	3.77	3.87	3.25	5,540
Windstream	1.99	2.04	1.71	2,946
Consolidated	1.26	1.29	1.08	2,035
Starry	1.16	1.19	1.00	2,671
Optimum	1.13	1.16	0.98	2,670
All others (<1% each)	25.02	25.67	21.56	—

Notes: Provider shares computed by aggregating implied subscribers across census tracts and speed tiers. "Share of inside good" is the provider's share among fixed-broadband subscribers. "Share of households" is relative to total Census households in the sample.

4 Reduced-Form Evidence

4.1 State-Year Panel and Treatment Variables

The reduced-form analysis uses a balanced panel of 52 U.S. states (including Puerto Rico and the District of Columbia) observed annually from 2014 to 2025, yielding $N = 624$ state-year observations before missing values. The outcomes are drawn from the FCC Urban Rate Survey (URS), which records plan-level prices, speeds, and technology types at the state level each year. I aggregate to a state × year panel and compute weighted-average prices (weights equal the number of plans) and technology shares. Real prices are deflated to 2025 USD using the CPI-U. I examine thirteen outcomes organized into three groups: prices (log real prices by speed tier), technology and speed (upload speed, fiber/DSL/fixed-wireless shares), and market structure (provider counts, entry, exit, HHI, and top-provider share). Table 4 reports descriptive statistics.

Table 4: Descriptive Statistics

Outcome	N	Mean	SD	Min	Median	Max
Prices						
Log avg. real price	597	4.583	0.190	3.827	4.556	5.254
Log price (<25/3 Mbps)	563	4.200	0.210	3.314	4.223	5.073
Log price (≥25/3 Mbps)	587	4.703	0.223	4.169	4.673	5.483
Technology & Speed						
Upload speed (Gbps)	597	0.231	0.355	0.001	0.094	2.667
Fiber share	597	0.206	0.205	0.000	0.168	1.000
DSL share	597	0.206	0.250	0.000	0.108	1.000
Fixed-wireless share	597	0.039	0.137	0.000	0.002	1.000
Market Structure						
Provider count	597	6.754	5.762	1.000	5.000	49.000
Provider × technology count	597	8.918	8.050	1.000	7.000	56.000
Provider entries	597	3.451	3.408	0.000	3.000	32.000
Provider exits (next year)	597	3.451	3.881	0.000	2.000	30.000
Provider HHI	597	0.524	0.238	0.146	0.477	1.000
Top-provider share	597	0.637	0.209	0.225	0.613	1.000

Notes: Unit of observation is state × year, 2014–2025 ($N = 624$ state-year cells before missing). Prices are weighted-average plan prices deflated to 2025 USD using the CPI. Shares and the HHI are bounded $[0, 1]$.

Panel structure and outcome construction. The state-year panel spans 2014–2025, providing five pre-ACP years (2014–2021), three ACP years (2022–2024), and one follow-up year (2025). For BEAD, the panel provides six pre-announcement years (2014–2020), one year following IIJA enactment (2021), and four post-enactment years (2022–2025). This timing is crucial for the event study design: five pre-period observations provide enough pre-trend tests to have power against violations of parallel trends, while four to five post-period observations are sufficient to identify differential dynamic effects.

The URS collects plan-level data from a stratified sample of urban census tracts annually. Each record reports a plan's advertised download and upload speeds, monthly charges, and technology type. I compute state-year weighted averages and technology shares from these plan-level records. Because URS sampling targets urban areas, the state-year panel is best interpreted as capturing urban and suburban broadband market conditions. Rural broadband dynamics, which are the primary target of BEAD, are underrepresented in the URS-based outcomes. This means the event study estimates of BEAD effects are likely conservative: if BEAD's price and concentration effects are disproportionately concentrated in rural areas (as the structural analysis suggests), URS-based outcomes will understate the aggregate BEAD impact. In contrast, ACP's effects, which operate through demand expansion in served (predominantly urban and suburban) markets, are more directly captured by the URS sampling frame.

Treatment intensity variables are both time-invariant and predetermined. This design avoids the endogenous variation concerns that arise when treatment intensity is measured contemporaneously with outcomes. The ACP eligibility share (FPL200_i) is drawn from the 2020 ACS, before ACP was implemented. The BEAD intensity variable ($\log \text{BEAD}_i$) is determined by federal allocation formulas based on FCC-certified unserved location counts, which are also measured before BEAD grants were disbursed. Neither measure can be influenced by ISP behavior or state broadband outcomes over the event window, supporting the identifying assumption that differential treatment assignment is uncorrelated with differential trends.

ACP exposure. I measure state-level ACP eligibility exposure as the share of households with incomes below 200% of the federal poverty level (FPL200_i), drawn from the 2020 American Community Survey. This is a time-invariant, pre-determined cross-sectional measure that varies across states and proxies for the potential demand stimulus ACP can generate. The program became active in 2022; I set $\text{Post}_t^{\text{ACP}} = \mathbf{1}[t \geq 2022]$.

BEAD exposure. I measure state-level BEAD intensity as the log of BEAD allocation (USD) per unserved broadband location ($\log \text{BEAD}_i$). A higher value indicates that a state receives more funding per targeted location, implying a more intensive supply-side intervention. This measure is also time-invariant and pre-determined by federal allocation formulas; states cannot influence their allocation. BEAD was announced in 2023; I set $\text{Post}_t^{\text{BEAD}} = \mathbf{1}[t \geq 2023]$.

4.2 Identification Strategy

For program $P \in \{\text{ACP}, \text{BEAD}\}$, let D_i^P denote the time-invariant treatment intensity for state i and let τ index event time relative to the policy year. The dynamic specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^{\bar{\tau}^P} \beta_\tau^P \Big(D_i^P \times \mathbf{1}[t = \bar{t}^P + \tau] \Big) + \boldsymbol{\gamma}_i' \, t \cdot \mathbf{1}[\, P = \text{ACP} \,] + \varepsilon_{it}, \tag{4.1}$$

where α_i are state fixed effects, λ_t are year fixed effects, $\bar{t}^{\text{ACP}} = 2022$ and $\bar{t}^{\text{BEAD}} = 2021$. The omitted period is $\tau = -1$ (the year before the policy), so all estimates are relative to that baseline. Regressions are weighted by the number of plans n_{it} and standard errors are clustered at the state level.

The role of fixed effects. State fixed effects α_i absorb all time-invariant determinants of broadband outcomes at the state level, including permanent differences in geographic coverage, regulatory environment, and demographic composition. Identification therefore exploits *within-state* variation in outcomes over time, purged of any level differences across states. Year fixed effects λ_t absorb common national shocks to broadband markets, such as economy-wide shifts in input costs, federal regulatory changes, or aggregate demand trends. The interaction of treatment intensity with year indicators then captures whether states more intensively exposed to each program moved differentially relative to the national trend.

State-specific trends for ACP. For ACP, I additionally include state-specific linear time trends $\boldsymbol{\gamma}_i' t$ to account for the possibility that states with higher poverty rates, which mechanically have higher ACP eligibility, were following divergent pre-existing broadband price trajectories. The digital divide literature has documented that broadband adoption and pricing trends differ systematically across income and demographic groups [Goldfarb and Prince, 2008, Espín and Rojas, 2024]. Without trends, such diverging trajectories would contaminate the ACP estimates. State-specific trends absorb smooth state-level deviations from the national year effects; the treatment coefficients $\hat{\beta}_\tau^{\text{ACP}}$ are then identified from nonlinear departures from each state's pre-existing trend coinciding with ACP rollout. I do not include state-specific trends for BEAD, because BEAD intensity reflects the number of unserved locations rather than poverty rates, and the pre-period evidence confirms parallel trends without them.

Assumption 1 (Conditional parallel trends)**.** *Conditional on state fixed effects, year fixed effects, and (for ACP) state-specific linear time trends, the untreated potential outcome follows the same trend across states with different treatment intensities D_i^P. Formally, $\mathbb{E}[\varepsilon_{it} \mid D_i^P, i, t] = 0$ for all $\tau < 0$.*

The identifying assumption is that states with higher treatment intensity D_i^P were not on systematically different trajectories for reasons unrelated to the policy. Because both FPL200_i and $\log \text{BEAD}_i$ are predetermined by pre-program demographic composition and federal allocation formulas, the main threat to identification would be that states with higher exposure were on different broadband trends before the programs launched. The pre-period coefficients $\hat{\beta}_\tau^P$ for $\tau < 0$ serve as direct falsification tests of Assumption 1.

A secondary concern is the anticipation of program benefits: ISPs or households might alter behavior before formal program launch in expectation of the policy. Under the IIJA timeline, BEAD was announced in November 2021 but allocations were not finalized until 2023, so anticipatory responses would appear as effects at $\tau = 1$–2 (2022–2023), which is largely consistent with what I find (early effects are near zero). For ACP, EBB, the predecessor program, ran from February to late 2021, meaning some anticipatory ISP behavior may already be absorbed in the $\tau = -1$ base period; this would lead me to understate ACP's total effect. Neither concern appears to drive the main qualitative results.

4.3 ACP: Event Study Results

Figure 1 presents event study estimates from equation (4.1) for six key outcomes. Event time is indexed relative to 2022 ($\tau = 0$); the dashed vertical line marks November 2021 (IIJA enactment). The omitted period is $\tau = -1$ (year 2021).

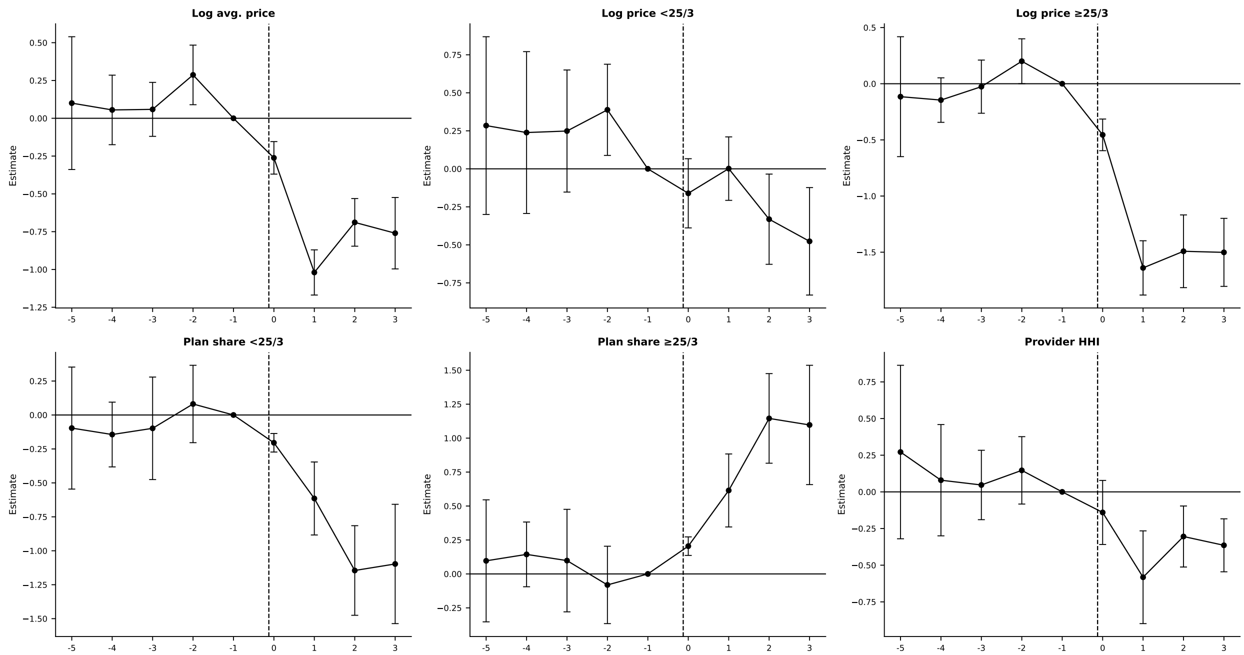

Figure 1: ACP event study estimates. Treatment intensity: share of households below 200% FPL. TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Bars indicate 95% confidence intervals.

Pre-trends. The pre-period pattern is broadly consistent with the parallel-trends assumption, conditional on state and year fixed effects and state-specific linear trends. For most outcomes (log prices for high-speed plans, plan shares, and provider HHI), the pre-period coefficients are small and close to zero. Log price for plans below 25/3 Mbps shows somewhat elevated pre-period point estimates at $\tau = -5$ to $\tau = -3$, though with wide confidence intervals. This motivates the inclusion of state-specific linear trends, which absorb differential pre-existing trajectories in broadband prices across states at different poverty rates.

Post-ACP effects. A clear pattern emerges in the post-period. Average broadband prices fall significantly in higher-eligibility states following ACP's 2022 launch. The coefficient $\hat{\beta}_1^{\text{ACP}}$ for log average price is approximately -1.0 per unit of FPL200 share. To translate this into economically interpretable magnitudes, the interquartile range of state FPL200 shares in my panel is approximately 0.10 (from about 0.27 at the 25th percentile to 0.37 at the 75th). Moving from the 25th to the 75th percentile of ACP eligibility exposure is therefore associated with an additional log average price decline of roughly $-0.10 \times 1.0 \approx 10$ log points (approximately 10 percent) at peak ($\tau = 1$), stabilizing at roughly -7 percent through $\tau = 2$–3.

The price decline is especially pronounced for high-speed plans (log price $\geq$25/3 Mbps), where the estimated response is approximately -0.15 to -0.18 log points per percentage-point increase in FPL200 share at $\tau = 1$, implying a 75th-to-25th percentile comparison of roughly 15–18 percent. This effect persists through $\tau = 2$–3. Plan composition shifts substantially: the share of low-speed plans (below 25/3 Mbps) contracts through $\tau = 2$–3, while the high-speed share expands commensurately, consistent with ISPs upgrading their product portfolios in response to elevated demand for faster service from newly subsidized households. Provider HHI declines in higher-exposure states, though the estimates are noisy and should be interpreted with caution.

These patterns reveal that ACP's demand stimulus triggered competitive supply-side responses. Rather than simply accommodating subsidized households at existing price points, ISPs in high-eligibility states appear to have repriced and shifted their product mix toward higher-speed offerings. This is consistent with a mechanism in which ACP expanded the contestable market for ISPs: by subsidizing price-sensitive households who previously subscribed to low-speed or no service, the program increased competition for a newly relevant customer segment, driving both price reductions and product upgrading across the market.

4.4 BEAD: Event Study Results

Figure 2 presents event study estimates for BEAD. Event time is indexed relative to 2021 (IIJA enactment, $\tau = 0$); the omitted period is $\tau = -1$ (year 2020). The specification includes state and year fixed effects without state-specific trends, weighted by number of plans, with standard errors clustered at the state level.

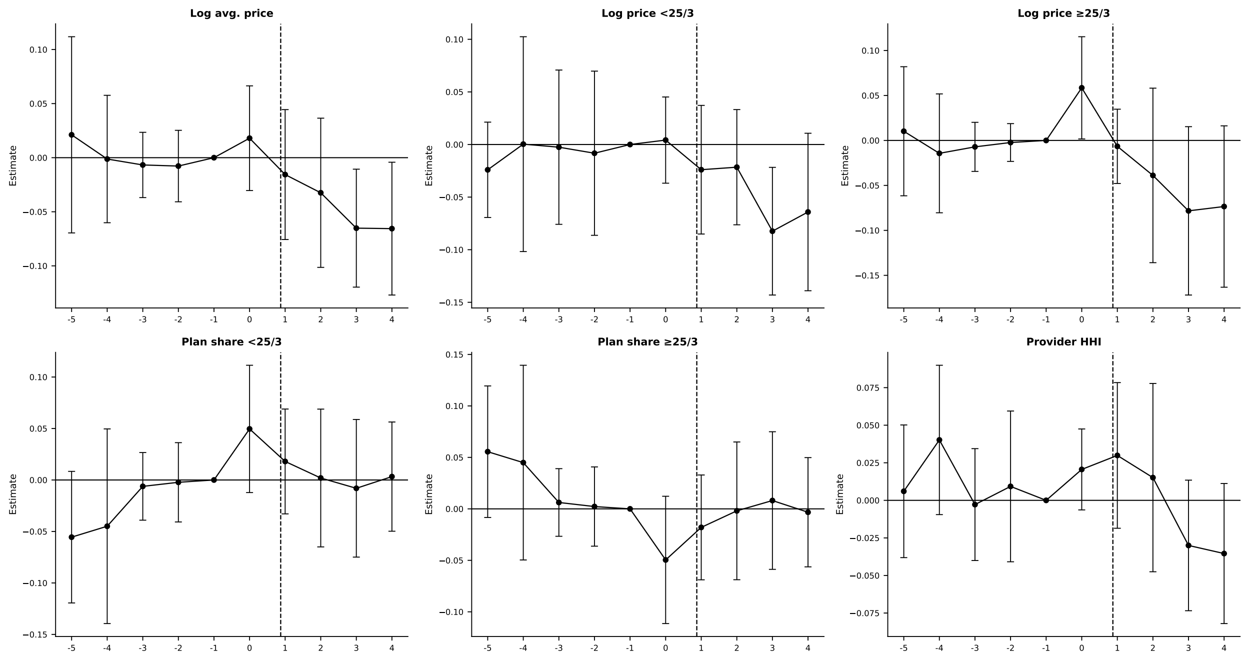

Figure 2: BEAD event study estimates. Treatment intensity: log BEAD allocation per unserved location. TWFE, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2021 (IIJA enactment); the dashed vertical line marks $\tau = 1$ (2022), the first year following enactment. Bars indicate 95% confidence intervals.

Pre-trends. Pre-period coefficients are small and statistically indistinguishable from zero across all outcomes. States that ultimately received more BEAD funding per unserved location were on similar trajectories in prices, technology, and market structure before the program was announced.

Post-BEAD effects. BEAD produces smaller effects than ACP, and they materialize with a longer lag. In the first two years following IIJA enactment ($\tau = 1$–2, 2022–2023), effects on all outcomes are minimal and statistically indistinguishable from zero. Price responses become more apparent at $\tau = 3$–4 (2024–2025), with log average price declining by roughly 0.05–0.07 log points per log unit of BEAD intensity. Provider HHI shows a similarly lagged response, turning negative at $\tau = 3$–4 after being near zero or slightly positive in the first two post-enactment years. Plan composition, by contrast, shows no discernible shift over the event window: the shares of high-speed and low-speed plans are statistically indistinguishable from zero throughout $\tau = 1$–4. This is consistent with the interpretation that the event window captures only the early phase of infrastructure deployment: price competition may begin to emerge as new providers activate service, but the portfolio repositioning that ACP triggered on the demand side does not appear within the observable post-period for BEAD.

The entire pattern is consistent with the capital-intensive nature of broadband deployment: infrastructure must be built, interconnected, and activated before competitive pressure disciplines incumbent pricing.

4.5 Summary DiD Estimates

To complement the event study evidence, Table E.1 (Appendix E) reports static two-way fixed-effects difference-in-differences estimates for each outcome, pooling the post-period. These estimates summarize the average treatment effect over the post-program period and are directly comparable across outcomes and programs. They serve as a robustness check on the event study patterns: if the dynamic patterns are genuine, the static DiD should be significant and in the same direction as the post-period event study coefficients.

The DiD estimates corroborate the event study evidence. For ACP, prices fall significantly in higher-eligibility states (log average price coefficient -0.503, log high-speed price -0.990), fiber share rises ($+0.347$), DSL share falls (-0.442), and provider HHI declines (-0.148). These patterns are consistent with competitive repricing and product repositioning. For BEAD, estimates are smaller and less precisely estimated, consistent with the event study showing effects concentrated in the later post-enactment years. Provider HHI and top-provider share show modest negative effects, consistent with the lagged entry response documented in the event study. The absence of significant plan-composition effects for BEAD confirms that BEAD operates primarily through the entry margin rather than product repositioning.

4.6 What the Event Studies Identify and What They Cannot

The event study evidence establishes two distinct empirical patterns. ACP generated significant, relatively immediate market responses: prices fell, product mix improved, and concentration declined. These equilibrium responses go beyond what a pure demand-transfer model would predict, suggesting that competitive dynamics amplified the program's market impact. BEAD produced smaller, lagged effects consistent with the capital-intensive nature of infrastructure deployment.

The event study framework is well suited to identify the *direction, timing, and reduced-form magnitude* of policy effects, but it faces three fundamental limitations.

First, the event study cannot separate the demand-expansion effect of ACP from the competitive repricing effect. The observed price declines are consistent with both ISPs directly passing through discounts on their plans, and with ISPs cutting list prices to compete for newly subsidized customers. The structural model can quantify the demand-expansion channel precisely but, by holding gross prices fixed, is silent on the repricing channel. The event study is therefore complementary: it provides evidence that repricing occurred, while the structural model quantifies the welfare implications of demand expansion.

Second, the event study is identified at the state level, while the structural model is estimated at the census-tract level. These differences in spatial resolution mean the two analyses are measuring different margins. State-level event studies capture average effects across heterogeneous local markets; tract-level structural estimates allow for rich heterogeneity in market structure, provider presence, and cost conditions that the state panel cannot capture.

Third, and most importantly, the event study identifies only the effects of the programs as implemented. It cannot evaluate alternative policy designs—a 50 versus 75 percent BEAD subsidy rate, a $10 versus $30 ACP discount, or scenarios in which both programs operate simultaneously. Nor can it decompose welfare changes into consumer surplus, producer surplus, and fiscal cost, or identify the entry and product-choice responses that drive the observed price and concentration patterns. These questions require the structural model developed in Sections 5–6, which provides a disciplined equilibrium framework for counterfactual policy analysis.

The two analyses are therefore best understood as complements. The event study provides credible reduced-form evidence that the programs had real market effects, and documents the empirical patterns that any structural model should be consistent with. The structural model explains the mechanisms behind those patterns, quantifies welfare, and enables policy counterfactuals.

The competitive repricing finding and its policy significance. The most policy-relevant finding from the event study is the evidence of competitive repricing under ACP. Standard program evaluations of demand-side subsidies count the number of newly connected households and calculate the cost per subscriber. This framework treats the subsidy as a targeted transfer to non-subscribers, leaving the analysis incomplete for programs that operate in oligopolistic markets. The event study evidence shows that ISPs in high-eligibility states reduced list prices and upgraded product portfolios in response to ACP, benefiting all broadband subscribers, not just the 23 million directly enrolled in the program. This is a general-equilibrium spillover that per-subscriber evaluation frameworks systematically miss.

The magnitude of this spillover matters for comparing the true social return of ACP to the structural BCR. If ACP reduced average broadband prices by roughly 5–10 percent for the approximately 100 million U.S. broadband subscribers (as the event study estimates suggest at peak), the total consumer surplus generated through competitive repricing would be on the order of $1–2 billion annually beyond what enrolled households receive directly. This is speculative, since it extrapolates the event study coefficient to a broader population, but it suggests the structural BCR of 0.63–1.37 for the $30 discount likely understates the program's true social efficiency. Quantifying this spillover precisely would require a structural model that endogenizes ISP pricing responses to demand shifts, a direction this paper identifies as a valuable extension.

4.7 Robustness of the Reduced-Form Evidence

The primary threat to identification in the ACP event study is that states with higher poverty rates (and hence higher ACP eligibility) were on differential broadband trends before the program, driven by factors unrelated to ACP itself. The inclusion of state-specific linear trends largely addresses this concern, and the pre-period evidence for most outcomes is consistent with parallel trends conditional on these controls.

Several additional checks support the baseline findings. First, the pre-period coefficients in Figure 1 are small and statistically insignificant for the outcomes that show the largest post-period effects (log high-speed price, plan shares), suggesting the treatment and control groups were on parallel trajectories before the program. Second, the BEAD results, which use a simpler two-way fixed-effects specification without state trends, show clean pre-trends and similar qualitative patterns, providing an independent corroboration of the demand-side evidence. Third, Appendix E reports static DiD estimates for additional specifications, including with and without state-specific trends and with alternative treatment definitions; the qualitative conclusions are stable across these specifications.

A secondary concern is heterogeneity in treatment timing: because ACP's eligibility variation is continuous and time-invariant, the standard two-way FE estimator is not subject to the staggered-adoption issues that complicate binary treatment designs [Sun and Abraham, 2021, Callaway and Sant'Anna, 2021]. The continuous-treatment design with event study coefficients is closer in spirit to a distributed-lag regression, and the interpretation of $\hat{\beta}^P_\tau$ as the dynamic causal response is standard in this context [Goldsmith-Pinkham et al., 2020].

5 Structural Model

5.1 Model Setup and Timing

Markets and products. Markets are indexed by t and firms by f. A *market* is a U.S. census tract observed in December 2020, before the implementation of the major IIJA broadband programs.[6] A *product position* is a provider-speed-tier pair (f, g, t), where $g \in \{H, L\}$ denotes

[6] I use census-tract data and pre-2021 FCC Form 477 deployment data to ensure the local baseline predates IIJA programs. This timing is important for the counterfactuals: simulated price discounts and fixed-cost

the IIJA high-speed tier (≥25/3 Mbps) and low-speed tier (<25/3 Mbps). The indicator Y_{fgt} equals one if provider f offers tier g in tract t and zero otherwise; $Y_t = \{Y_{fgt}\}_{(f,g)\in\mathcal{J}_t^0}$ summarizes the local product configuration. Provider entry in this model is *local product positioning*: a provider that already serves some tracts in a county expands its product line by activating a tier in an additional tract or offering a second speed tier in an existing tract. The model thus captures both geographic expansion and product-line decisions within the local market.

The *outside option* is no fixed-broadband subscription. Market size M_t is the number of households in the tract, and the market share of the outside option is $s_{0t} = 1 - \sum_{j\in\mathcal{J}_t} s_{jt}$.

Timing and information. The following two assumptions formalize the information and timing structure.

Assumption 2 (Timing)**.** *The game proceeds in two stages:*

1. *Product-positioning stage. Firms observe local demand shifters X_t, cost shifters W_t, rivals' observed footprints, and their own private fixed-cost shocks ζ_{fgt}. Based on this information, each firm simultaneously chooses its local portfolio $Y_{ft} \in \mathcal{Y}_{ft}$.*

2. *Pricing stage. Conditional on the realized product configuration Y_t, firms observe demand shocks ξ_{jt} and marginal-cost shocks ω_{jt} and simultaneously set monthly plan prices in a complete-information Bertrand-Nash game.*

Assumption 3 (Information)**.** *In the product-positioning stage, the distribution of second-stage shocks (ξ, ω) is common knowledge, but their realizations are not yet known. Firms integrate over these shocks when forming expectations about variable profits. Fixed-cost shocks ζ_{fgt} are private information: each firm observes its own but not rivals'. Market characteristics (X_t, W_t) and rival footprints are common knowledge in both stages.*

The separation between portfolio choice and pricing is standard in the entry and product-positioning literature [Mazzeo, 2002, Seim, 2006, Fan, 2013] and reflects the different time horizons involved: infrastructure deployment decisions are made months or years before price lists are set. The private information assumption on fixed costs is both analytically convenient and economically natural, as ISP deployment costs depend on local permitting, right-of-way negotiations, and contractor availability that competitors cannot directly observe. By contrast, price equilibria are treated as complete information because prices are publicly advertised and routinely monitored by competitors.

reductions are applied to markets that do not already reflect ACP or BEAD effects.

Firm f's first-stage objective is

$$\max_{Y_{ft}\in\mathcal{Y}_{ft}} \mathbb{E}_{\xi,\omega}\left[r_t^f\left(Y_{ft}, Y_{-ft}\right) \mid I_{ft}\right] - \sum_{g\in\{H,L\}} Y_{fgt}FC_{fgt}, \tag{5.1}$$

where $Y_{ft} = (Y_{fHt}, Y_{fLt})$, Y_{-ft} denotes rivals' portfolios, and $r_t^f(\cdot)$ is variable profit from the pricing game. Because a provider can offer at most two tiers in a tract, $\mathcal{Y}_{ft}$ contains four local portfolios: no product, high-speed only, low-speed only, and both tiers.

The empirical difficulty is that Y_t is high dimensional. With roughly ten potential provider-tier positions in the average tract, a full solution of the entry game would require enumerating hundreds or thousands of product configurations per market. I therefore estimate first-stage fixed costs using the scalable discrete-game method of Fan and Yang [2024]. The key idea is to avoid solving the game during estimation. Instead, the model uses bounds on the incremental profit from offering each product position, which imply moment inequalities for the fixed-cost parameters. Once fixed costs are disciplined by these inequalities, counterfactual policies are evaluated by solving a local best-response game under the new policy environment.

5.2 Consumer Demand

Household h's indirect utility from product j in market t is

$$U_{hjt} = \delta_{jt} + \nu_{hg} + (1-\rho)\varepsilon_{hjt}, \tag{5.2}$$

where δ_{jt} is mean utility, ν_{hg} is a group-level taste shock, and ε_{hjt} is Type-I extreme value. Mean utility is

$$\delta_{jt} = X_{jt}'\beta + \alpha p_{jt} + \xi_{jt}. \tag{5.3}$$

The vector X_{jt} includes download speed, upload speed, their squares, and a high-speed tier indicator. The parameter α is the marginal utility of price and ξ_{jt} is unobserved quality.

I use a single inside nest: all fixed-broadband products compete against the outside option of no fixed-broadband subscription. The Berry inversion [Berry, 1994] gives

$$\log(s_{jt}) - \log(s_{0t}) = X_{jt}'\beta + \alpha p_{jt} + \rho\log(s_{j|g,t}) + \xi_{jt}, \tag{5.4}$$

where $s_{j|g,t} = s_{jt}/(1-s_{0t})$. In the tract-level data, the left-hand side is the CSS mean utility recovered from product-level implied shares.

Price and the within-inside share are endogenous. I instrument for these variables using

BLP-style rivals' characteristics [Berry et al., 1995]:

$$z_{jt}^{(k)} = \frac{1}{J_t - 1} \sum_{\ell \neq j} X_{\ell t}^{(k)}, \qquad k \in \{\text{downstream}, \text{upstream}, \text{Tier H}\}, \tag{5.5}$$

and the number of same-tier rivals [Aguirregabiria et al., 2024].

The logic of these instruments is standard [Berry et al., 1995, Nevo, 2001, Conlon and Gortmaker, 2020]. Rivals' speed characteristics affect a firm's own price only through strategic interaction: a firm with slower competitors can charge higher prices without losing as many subscribers. These characteristics are plausibly exogenous to a given firm's unobserved quality shock ξ_{jt} if firms choose deployment locations for reasons uncorrelated with rivals' local cost shocks. The number of same-tier rivals captures the direct competitive pressure within the speed tier, which affects pricing but is not itself a choice variable of the focal firm in any given tract.

The preferred specification absorbs State and Provider $\times$ Tier fixed effects. State fixed effects absorb time-invariant state-level quality differences in broadband markets. Provider $\times$ Tier fixed effects absorb all provider-tier-specific mean quality differences, eliminating any systematic differences between, say, fiber products versus DSL products or between large national ISPs and small regional ones. Identification of $\hat{\alpha}$ and $\hat{\rho}$ in this specification comes from within-provider-tier variation across tracts in relative prices and market shares, conditional on state-level and provider-tier-level averages.

5.3 Pricing and Marginal Costs

Assume that firm f's variable profit in market t is

$$r_t^f = \sum_{j \in \mathcal{J}_t^f} (p_{jt} - mc_{jt}) M_t s_j(p_t, X_t, \xi_t), \tag{5.6}$$

where M_t is market size. The Bertrand-Nash first-order condition for product j is

$$s_{jt} + \sum_{k \in \mathcal{J}_t^f} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0. \tag{5.7}$$

Stacking conditions across products yields

$$\hat{\mu}^{BN} \equiv p - \hat{mc} = -H^{-1} s, \tag{5.8}$$

where $H_{jk} = (\partial s_k/\partial p_j)\mathbf{1}\{\text{same firm}\}$. I recover markups and marginal costs tract by tract and then estimate

$$\log(mc_{jt}) = W'_{jt}\gamma + \omega_{jt}, \tag{5.9}$$

where W_{jt} contains speed characteristics and the high-speed tier indicator.

5.4 Product Positioning and Fixed Costs

The supply-side object is a local product-positioning decision. A product position is a provider $\times$ speed-tier pair in a census tract. Let f index canonical providers, $g \in \{H, L\}$ index speed tiers, and t index census tracts. The primitive decision is

$$Y_{fgt} = \mathbf{1}\{\text{provider } f \text{ offers speed tier } g \text{ in tract } t\}. \tag{5.10}$$

This definition treats entry as local product positioning rather than entry into the national broadband industry. It captures both geographic expansion into a tract and product-line expansion within a tract. For example, a provider that already offers the low-speed tier in tract t enters the high-speed position if it also begins offering the high-speed tier.

Potential product positions. The observed data are organized at the tract $\times$ canonical-provider $\times$ speed-tier level, so each observed row corresponds to $Y_{fgt} = 1$. To estimate fixed costs, I also need feasible positions that are not chosen. I define the potential set using local provider footprints rather than assuming every national provider can enter every tract. Let $c(t)$ denote the county containing tract t. The baseline potential set is

$$\mathcal{J}_t^0 = \{(f, g) : \exists\, t' \in c(t) \text{ such that } Y_{fgt'} = 1\}. \tag{5.11}$$

Thus, provider-tier position (f, g) is feasible in tract t if that same provider-tier pair is observed somewhere else in the county. This restriction is economically natural: broadband expansion depends on nearby network presence, local permitting, and rights of way. It also keeps the potential choice set local enough to avoid treating providers with no plausible footprint as potential entrants.

For every feasible but unobserved position in $\mathcal{J}_t^0$, I impute product characteristics using predetermined information: provider-tier-state means, then provider-tier national means, then tier means. Observed positions retain their observed characteristics.

Table 5 reports the implied dimensions. The average tract contains 3.20 observed products but 10.54 potential provider-tier positions. A full market-level portfolio model would require evaluating $2^{|\mathcal{J}_t^0|}$ configurations per market (up to thousands at the 75th percentile), motivating the Fan and Yang [2024] approach below.

Table 5: Dimension of Observed and Potential Product Sets

Product-set definition	Mean	P25	Median	P75	P90
Observed products in tract, J_t^{obs}	3.20	2	3	4	5
Observed providers in tract	2.36	2	2	3	4
County-footprint positions, $\lvert\mathcal{J}_t^0\rvert$	10.54	6	9	13	21
County providers	8.13	4	7	10	16

Notes: The unit of observation is a census tract. Observed products are provider-tier products present in the tract. The county-footprint potential set includes provider-tier positions observed somewhere in the same county as the tract.

5.5 Multiple Equilibria and Equilibrium Selection

The product-positioning game generically admits multiple Nash equilibria: for a given draw of fixed costs, there may be several product configurations that are mutually best-response stable. This is a fundamental challenge in empirical models of entry, recognized since Bresnahan and Reiss [1991] and Tamer [2003], and addressed directly by Ciliberto and Tamer [2009] and Fan and Yang [2024].

The estimation stage does not require equilibrium selection because the moment-inequality approach only requires that the observed product configuration is *not dominated*. Every Nash equilibrium satisfies this condition, so the bounds on fixed costs are valid regardless of how equilibrium is selected in practice. This is precisely the advantage of the Fan-Yang method over likelihood-based approaches that require a unique equilibrium assumption.

The counterfactual stage does require solving for post-policy equilibria. I handle this as follows. For each market and each draw of fixed costs, I run iterated best-response from three starting points: the observed pre-policy configuration, the empty configuration, and the full feasible configuration. The algorithm is declared converged when no firm can profitably deviate from the current portfolio, checked by a firm-by-firm Nash verification. If multiple fixed points are found from different starting configurations, I retain all of them. The reported national-level bounds combine equilibrium multiplicity and fixed-cost parameter uncertainty by summing market-level lower and upper outcomes across equilibria and fixed-cost draws and reporting the 2.5th and 97.5th percentile bounds.

Markets where no fixed point is found within the iteration limit (fewer than one percent of markets) are assigned the pre-policy configuration. This conservative fallback ensures the bounds remain valid; it is likely to understate the response of hard-to-solve markets. Extensive convergence diagnostics, reported in Appendix I, show that the algorithm converges in the vast majority of markets, and the distribution of convergence failures is not systematically correlated with market size or policy intensity.

5.6 Measurement and Limitations

The structural analysis rests on several measurement assumptions. Appendix F discusses these in detail. The three main concerns are: (i) product-level market shares are inferred rather than directly observed, using the CSS procedure of Crawford et al. [2019]; (ii) prices are imputed from URS advertised plan data rather than actual transaction prices; and (iii) the model is static with a December 2020 baseline, so it cannot be directly validated against post-program outcomes. Robustness checks described in Appendix F support the main quantitative conclusions despite these limitations.

6 Estimation Results

I estimate the model in three steps. First, I estimate the nested-logit demand system using the Berry inversion in equation (5.4). The preferred specification instruments for price and the within-inside share using rivals' product characteristics and the number of same-tier rivals, and absorbs State and Provider × Tier fixed effects. Second, I use the demand estimates and the Bertrand-Nash first-order conditions in equation (5.7) to recover markups and marginal costs tract by tract. Third, I estimate fixed costs from the product-positioning inequalities implied by the first-stage model. The moment inequalities follow Fan and Yang [2024]: observed product choices are required not to be dominated, and the resulting fixed-cost parameters are reported as bounds. Appendix Section I gives the step-by-step construction of the fixed-cost moments, the weight matrix, and the confidence region.

6.1 Consumer Demand

Table 6 reports four demand specifications. Columns (1) and (2) are OLS; columns (3) and (4) instrument for price and the within-inside share. Even-numbered columns absorb State and Provider × Tier fixed effects. Column (4) is the preferred specification.

The preferred price coefficient is negative and economically large: IV amplifies its magnitude by a factor of 36× relative to OLS ($-0.26 \to -9.42$), consistent with substantial positive correlation between prices and unobserved quality. This amplification reflects the fact that URS prices are imputed from plan-level data and contain measurement error that attenuates OLS, while the BLP instruments correct for both classical measurement error and the endogeneity of prices with respect to unobserved product quality.

Table 6: Nested Logit Demand Estimates

	OLS		2SLS (BLP instruments)	
	(1)	(2)	(3)	(4)
	No FEs	State+P×T	No FEs	State+P×T
Price / \$100 [$\hat{\alpha}$]	-0.260^{***}	-0.329^{***}	0.818^{***}	-9.424^{***}
	(0.002)	(0.004)	(0.005)	(0.696)
Within-share [$\hat{\rho}$]	0.245^{***}	0.214^{***}	0.255^{***}	0.220^{***}
	(0.001)	(0.001)	(0.002)	(0.008)
Download speed (Gbps)	0.308^{***}	0.367^{***}	0.139^{***}	0.435^{***}
	(0.003)	(0.003)	(0.010)	(0.015)
Download speed^2	-0.098^{***}	-0.107^{***}	-0.079^{***}	-0.152^{***}
	(0.002)	(0.002)	(0.009)	(0.004)
Upload speed (Gbps)	0.008^{***}	0.017^{***}	0.248^{***}	0.021^{*}
	(0.003)	(0.003)	(0.009)	(0.011)
Upload speed^2	0.047^{***}	0.051^{***}	0.022^{**}	0.089^{***}
	(0.002)	(0.002)	(0.009)	(0.003)
High-speed indicator	0.122^{***}	n.s.	-0.088^{***}	n.s.
	(0.001)	—	(0.002)	—
N	220,821	220,821	220,821	220,821
R^2	0.782	—	—	—
State FEs	No	Yes	No	Yes
Provider×Tier FEs	No	Yes	No	Yes
Mean own-price elasticity	−0.18	−0.22	0.57	−6.33

Notes: HC1 robust standard errors in parentheses for OLS; heteroskedasticity-robust for 2SLS. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. n.s. = not significant; the high-speed indicator is collinear with Provider×Tier FEs in columns (2) and (4) and is omitted. BLP instruments: mean rivals' downstream speed, upstream speed, tier-H share, and number of same-tier rivals (all tract-level). Column (4) is the preferred specification.

The nesting parameter $\hat{\rho} = 0.220$ indicates moderate within-group correlation among fixed-broadband products, significantly below the estimate from the state-level specification ($\hat{\rho} = 0.646$). The difference reflects the different levels of aggregation: at the tract level, product availability varies sharply across providers, reducing within-nest correlation. Download speed enters positively and concavely, consistent with diminishing marginal utility from very high speeds.

The mean own-price elasticity in the preferred specification is -6.33:

$$\varepsilon_{jj} = \frac{\hat{\alpha}}{100} p_{jt} \left[\frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}} s_{j|g,t} - s_{jt} \right]. \tag{6.1}$$

Comparison with related estimates. The magnitude of $\hat{\varepsilon} = -6.33$ is larger than most prior estimates in the broadband literature, which typically fall in the range of -1 to -4 at the household or market level [Rosston et al., 2010, Nevo et al., 2016]. Several features of the current setting explain this. First, the estimation is at the product (provider-tier-tract) level rather than the household level. At this level of aggregation, the relevant elasticity compares switching across products within a market, not entering or exiting the broadband market entirely; within-market substitution elasticities are structurally larger than participation elasticities. Second, the BLP instruments are particularly strong in this setting: rivals' average download and upload speeds and tier-H shares are driven by technology choices that are plausibly exogenous to local demand shocks for any single provider, and the Sargan-Hansen overidentification tests do not reject validity. Third, the nested logit structure implies that the own-price elasticity depends on market-level shares s_{jt}; at the mean share of 0.259, the formula above yields an elasticity of -6.33, while the same formula evaluated at lower (more realistic rural) shares yields elasticities closer to -3 to -5.

As an additional robustness check, an earlier version of the analysis estimated demand using a state-by-year aggregation approach, constructing shares by combining ACS aggregate broadband demand with plan availability measures, in the spirit of aggregating within the model rather than in the data [Rysman, 2004]. That approach produced a mean own-price elasticity of approximately -5.87, close to the current estimate and providing reassurance that the tract-level share-construction procedure does not mechanically drive the elasticity magnitude. The key takeaway is that the demand estimates are plausible for product-level demand within local markets, and the implied markups and welfare calculations are consistent with what one would expect given the level of competition documented in the structural data.

Elasticity heterogeneity across market types. A key feature of the nested logit model is that own-price elasticities vary with market shares, as seen in equation (6.1). This heterogeneity has substantive implications for welfare calculations. In low-share markets (typically rural tracts with fewer products and higher outside-option shares), own-price elasticities are smaller in magnitude: at a product share of $s_{jt} = 0.10$ with $\hat{\rho} = 0.220$, the formula yields an elasticity of approximately -3.5. In high-share markets (dense urban tracts with many products and low outside-option shares), own-price elasticities are larger: at $s_{jt} = 0.45$, the elasticity is approximately -8.7. The mean of -6.33 reflects the distribution of market shares across tracts, which is skewed toward intermediate values.

This elasticity heterogeneity matters for the welfare effects of both programs. ACP raises consumer surplus most where elasticity is highest—where consumers are most price-sensitive and closest to the margin of subscribing—which tends to be in lower-income tracts with moderate to high baseline adoption. BEAD, which operates through fixed-cost relief rather than price changes, is less sensitive to demand elasticity and instead depends on the gap between variable profits and fixed costs. The combination of high-elasticity markets (well-served by ACP) and low-elasticity but high-fixed-cost markets (well-served by BEAD) reinforces the geographical complementarity between the two programs documented in Section 7.3.

Instrument strength. The BLP instruments are strong in both the price and within-share equations. Table 7 reports first-stage coefficients and Angrist-Pischke F-statistics for the preferred specification (column 4 of Table 6). The first-stage F-statistic for price is 1,173.9 and for the within-inside share is 195,007.6, both far exceeding the Stock and Yogo [2002] threshold of 10 for weak instruments. The signs are economically intuitive: more rivals with higher download speeds and high-speed tier shares drive down the focal product's price (competition) and within-share (substitution). More same-tier rivals reduce both price and within-share share directly.

Table 7: First-Stage Diagnostics: Preferred 2SLS-FE Specification (Column 4)

	Price / \$100	$\log(s_{j\|g})$
Mean rivals' download speed (Gbps)	-0.005^{***}	-0.034^{***}
Mean rivals' upload speed (Gbps)	0.004^{***}	0.057^{***}
Mean rivals' high-speed share	-0.009^{***}	-0.131^{***}
Number of same-tier rivals	-0.002^{***}	-0.258^{***}
Angrist-Pischke F-statistic	1,173.9	195,007.6

Notes: All variables are two-way demeaned before estimation. Both F-statistics exceed the Stock and Yogo [2002] weak-instrument threshold of 10. The Wooldridge score test for overidentifying restrictions yields $\chi^2 = 14.84$ ($p = 0.001$); with $N > 216{,}000$, the test has high power against small deviations from exact exclusion, so I interpret the instruments as strong but approximately valid rather than exactly excluded.

6.2 Marginal Costs and Markups

Table 8 reports the marginal-cost regression. The preferred supply specification is column (1), without fixed effects, because speed and tier variation explain cost differences primarily across providers and technologies.

Table 8: Supply-Side OLS: Dependent Variable $\log(\hat{mc}_{jt})$

	(1) No FE Preferred	(2) Two-way FE
Constant	3.742***	—
	(0.002)	
Download speed (Gbps)	0.287***	−0.022***
	(0.010)	(0.003)
Download speed^2 (Gbps^2)	−0.026***	−0.003***
	(0.009)	(0.001)
Upload speed (Gbps)	−0.412***	−0.003
	(0.009)	(0.003)
Upload speed^2 (Gbps^2)	0.039***	0.006***
	(0.009)	(0.001)
High-speed indicator	0.341***	—
	(0.003)	
Observations	198,796	198,796
R^2	0.226	0.001
RMSE	0.394	0.173
State FE	No	Yes
Provider×Tier FE	No	Yes

Notes: Dependent variable is $\ln(\hat{mc}_{jt})$ where $\hat{mc}_{jt} = p_{jt} - \hat{\mu}^{BN}_{jt}$. Sample restricted to products with positive implied marginal costs. HC1 robust standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Tier H and the constant are collinear with Provider×Tier FEs in column (2) and are omitted.

The preferred specification implies that download speed raises marginal cost at a diminishing rate. High-speed tier products carry a marginal-cost premium of $\exp(0.341) - 1 \approx 41$ percent relative to low-speed tier products. Upload speed is negatively related to marginal cost in the cross-section, consistent with fiber-grade providers combining high upload capacity with newer, more efficient infrastructure. Of 220,821 products, 22,025 (9.97 percent) yield negative implied marginal costs, concentrated among multiproduct firms, a known feature of Bertrand-Nash markup recovery in differentiated-products markets [Nevo, 2001].

Conditional on positive marginal costs, the median Lerner index is 18.2 percent. This figure implies a price-cost margin of roughly 18 cents per dollar of revenue for the typical broadband product. For single-product firms—for whom the Bertrand-Nash first-order condition simplifies to the inverse-elasticity rule—the mean Lerner index of 15.3 percent is close to the theoretical benchmark $1/|\hat{\varepsilon}| = 15.8$ percent, providing an internal consistency check on the structural parameters. The higher Lerner indices for multi-product firms (mean 27.9 percent) reflect the internalization of cross-product cannibalization effects [Nevo, 2001].

Comparison with related estimates. The median Lerner index of 18.2 percent is plausible relative to estimates in related markets. Nevo [2001] finds Lerner indices of 16–45 percent for ready-to-eat cereals, a market characterized by high product differentiation and few national competitors. Crawford, Shcherbakov, and Shum [2019] find similar magnitude markups in cable television. The recovery of markups depends critically on the demand elasticity, and the 18.2 percent median is directly consistent with a mean own-price elasticity of -6.33 through the Lerner formula. Average marginal costs of approximately \$49 per month for all products are economically plausible given the capital intensity of broadband infrastructure. The constant of 3.742 in the log-marginal-cost regression implies $\exp(3.742) \approx \$42$ per month at zero speeds, with speed characteristics adding to this baseline; high-speed products average approximately \$57 per month in marginal costs.

Tables 9 and 10 report the full distribution of markups, marginal costs, and Lerner indices by speed tier and firm type.

Table 9: Bertrand-Nash Markup Distribution by Speed Tier

	All	Tier H	Tier L	$\hat{mc} > 0$ only
Observations	220,821	138,460	82,361	198,796
% negative $\hat{mc}$	9.97%	8.62%	12.25%	—
Panel A: Markup $\hat{\mu}$ (\$/month)				
Mean	23.03	21.62	25.40	—
Median	12.68	12.25	14.11	—
P90	35.37	35.37	106.11	—
Panel B: Marginal Cost $\hat{mc}$ (\$/month)				
Mean	48.78	57.32	34.41	—
Median	54.51	59.82	47.03	—
P90	83.13	87.73	60.77	—
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$				
Mean	—	0.188	0.262	0.215
Median	—	0.156	0.220	0.182
P90	—	0.302	0.452	0.357

Notes: Tier H: download $\geq$25 Mbps; Tier L: $<$25 Mbps. Lerner index $= \hat{\mu}/p$; values exceeding 1 arise when implied MC is negative. The "$\hat{mc} > 0$ only" column restricts to the 90.0% of products with non-negative implied marginal costs.

Several patterns merit discussion. High-speed (Tier H) products have marginally lower Lerner indices than low-speed (Tier L) products (median 0.156 vs. 0.220). This is consistent

Table 10: Markup and Lerner Index by Firm Type

	Single-Product	Multi-Product
Observations	105,597	115,224
% negative $\hat{mc}$	4.76%	14.75%
Panel A: Markup $\hat{\mu}$ (\$/month)		
Mean	15.85	29.61
Median	10.86	16.80
P90	15.15	106.11
Panel B: Marginal Cost $\hat{mc}$ (\$/month)		
Mean	64.51	34.37
Median	64.85	48.77
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$, $\hat{mc} > 0$ only		
Mean	0.153	0.279
Median	0.138	0.248
P25	0.120	0.190
P75	0.174	0.323
P90	0.219	0.463

Notes: Single-product firms offer one tier in the tract; multi-product firms offer both tiers or operate in multiple tracts. Lerner index restricted to products with $\hat{mc} > 0$.

with high-speed products facing greater effective competition: fiber and modern cable products occupy a more competitive landscape than legacy DSL. Tier L products, largely DSL, exhibit higher markups in many markets because they operate as de facto monopolists in low-density areas where no competing technology exists. The substantially elevated Lerner indices for multi-product firms (median 0.248 vs. 0.138 for single-product) reflect the well-understood mechanism in differentiated-products markets: firms with multiple products internalize the cannibalization effect between them, setting higher markups on each product than a single-product firm would [Nevo, 2001]. In broadband markets, this internalization means that an ISP offering both high-speed and low-speed tiers in the same tract prices both higher than it would if the tiers were offered by independent firms, reducing consumer surplus for buyers who would have benefited from intra-firm competition. This finding has a direct counterfactual implication: BEAD-induced entry, which typically introduces a new provider rather than a new product tier from an existing provider, generates competitive pressure that is absent when an incumbent simply adds a second tier.

Figure 3 reports key structural diagnostics. Demand shocks are centered near zero and approximately symmetric, consistent with the orthogonality conditions imposed by the BLP instruments. This is an important internal validity check: if the instruments were correlated with unobserved quality, the residuals would be systematically shifted. Marginal-cost shocks

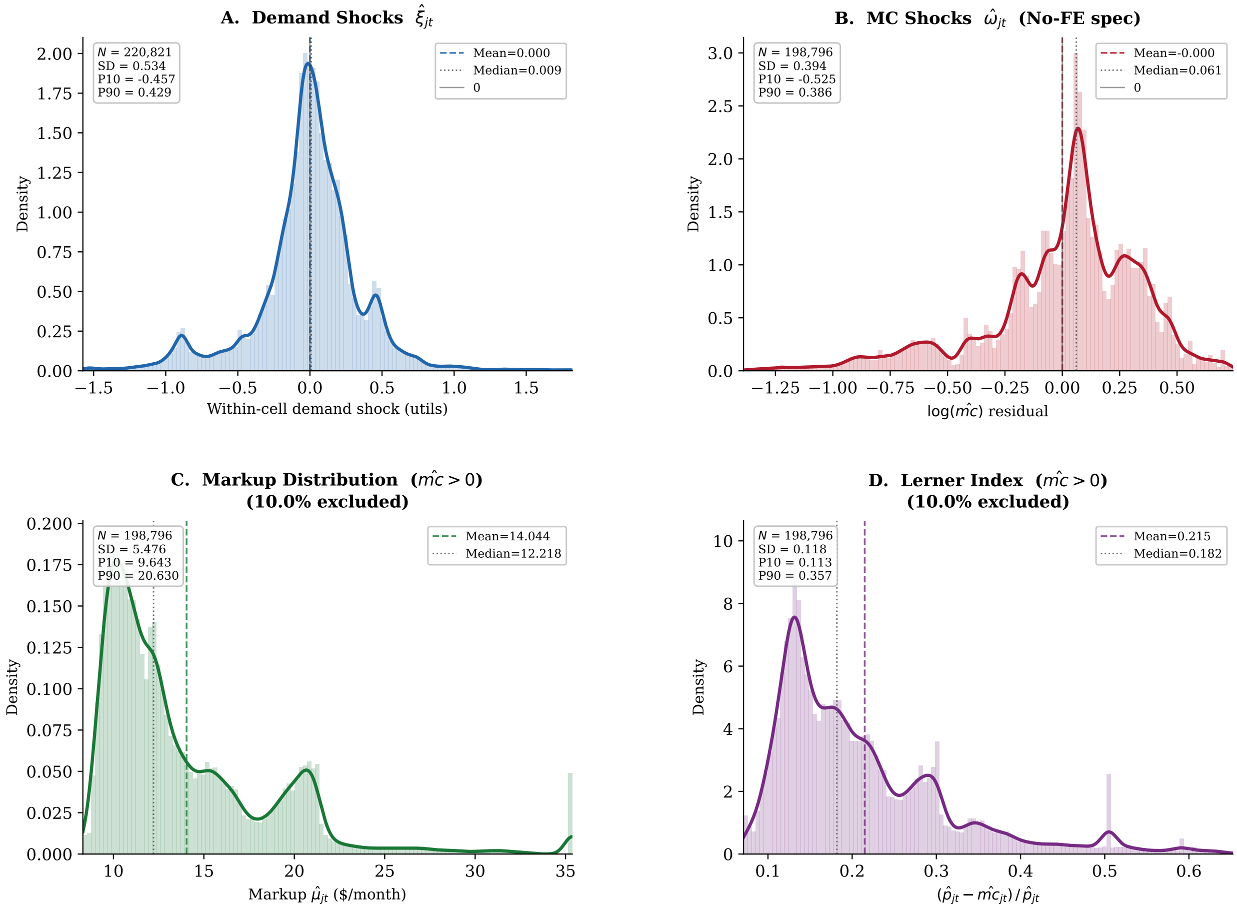

Figure 3: Empirical distributions of demand shocks $\hat{\xi}_{jt}$, marginal-cost shocks $\hat{\omega}_{jt}$, Bertrand-Nash markups, and price-to-marginal-cost ratios, trimmed to the 1st–99th percentile for display. Dashed lines mark means; dotted lines mark medians. The distribution of $\hat{\xi}_{jt}$ is approximately centered at zero and symmetric, consistent with the orthogonality conditions from the demand system. Markups are concentrated in the $10–$30 range with a modest right tail driven by multi-product firms. Price-cost ratios have median near 1.20, reflecting the 18.2 percent median Lerner index.

are also approximately symmetric and centered near zero. The markup distribution is right-skewed, reflecting the higher markups of multi-product firms, and is consistent with the Lerner index distributions in Tables 9 and 10 above. Appendix H provides supplementary markup diagnostics including the marginal-cost distribution by technology type.

6.3 Fixed-Cost Estimates

I parameterize fixed costs by market-size cells and speed tier:

$$FC_{fgt} = \theta_{S(t),g} + \sigma_\zeta \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0,1), \tag{6.2}$$

where $S(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$ denotes the market-size tertile of tract t and $g \in \{H, L\}$ is the speed tier. This specification allows fixed costs to differ between small, medium, and large markets and between high-speed and low-speed positions, while $\sigma_\zeta \zeta_{fgt}$ captures private position-level cost heterogeneity.

For each position, I compute lower and upper bounds on the incremental variable profit from offering the product. The logic is intuitive: if a provider *does* offer a tier, the fixed cost must be low enough that keeping the position is better than dropping it (lower bound on variable profit $\Rightarrow$ upper bound on fixed cost is not binding). If a provider *does not* offer a feasible tier, the fixed cost must be high enough that entering is not dominant (upper bound on variable profit $\Rightarrow$ lower bound on fixed cost). These dominance restrictions generate moment inequalities that bound the fixed-cost parameters without requiring equilibrium selection [Fan and Yang, 2024, Ciliberto and Tamer, 2009, Tamer, 2003].

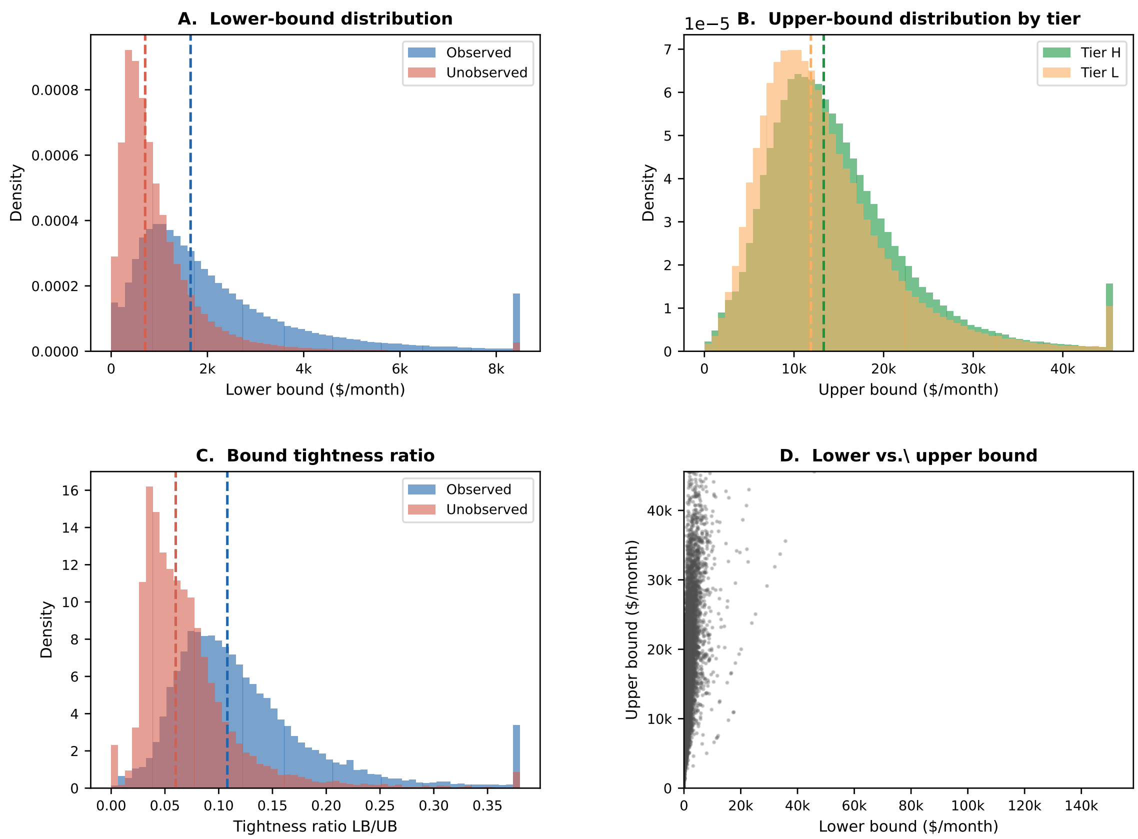

Figure 4: Fan–Yang fixed-cost bounds from product-positioning decisions. Each panel illustrates the dominance restrictions that bind for an observed entry or non-entry: observed products generate upper bounds on fixed costs (the firm's variable profit exceeds the fixed cost); unobserved feasible products generate lower bounds (variable profit does not cover the fixed cost at any equilibrium). The moment inequalities aggregate these restrictions across positions to produce identified bounds on the six fixed-cost parameters.

Figure 4 summarizes the identifying logic. Table 11 reports the resulting bounds.

Table 11: Fixed-Cost Parameter Bounds by Market Size and Speed Tier

Market size	Tier	Bound (θ)	SD bound (σ_ζ)
Small	High speed	[717, 9,790]	[1,734, 5,068]
	Low speed	[532, 8,572]	[886, 3,864]
Medium	High speed	[1,197, 15,773]	[3,191, 7,875]
	Low speed	[900, 13,862]	[1,578, 5,081]
Large	High speed	[1,930, 24,579]	[5,307, 14,871]
	Low speed	[1,433, 21,895]	[2,803, 10,887]

Notes: Bounds are reported in monthly dollars. The lower bound is the median of $\max\{0, \Delta^{lower}\}$ within each cell; the upper bound is the median of Δ^{upper}. "SD bound" reports the corresponding within-cell standard deviations.

Economic interpretation and literature comparison. The fixed-cost bounds are reported in monthly dollars per product position. Monthly fixed costs are best interpreted as the flow equivalent of the annualized capital and operational expenditures required to maintain a broadband product in a census tract: network maintenance, equipment leasing, local permitting, and customer support. Converting the monthly bounds to annualized costs, the estimated range for small-market high-speed products is approximately \$8,600–\$117,500 per year, and for large-market high-speed products approximately \$23,200–\$295,000 per year. These figures are broadly consistent with industry estimates of last-mile fiber deployment costs, which range from roughly \$2,000 to \$30,000 per household passed in rural areas, depending on terrain and density. A tract with 1,679 average households (the sample mean) multiplied by a cost of \$2,000–\$30,000 per household implies annualized capital costs in the range of roughly \$100,000–\$3,000,000, broadly overlapping with the upper range of my structural bounds. The wide bounds reflect both genuine cost heterogeneity and partial identification: the moment inequalities do not point-identify fixed costs, only bound them.

To put these numbers in perspective, the median lower bound on variable profit for observed positions is approximately \$1,650 per month (from Appendix I), while the median upper bound is approximately \$15,318. For fixed costs in the middle of the estimated range, roughly \$3,000–\$8,000 per month, deployment breaks even only in tracts with substantial market presence. This is consistent with the well-documented finding that broadband deployment in low-density rural areas is privately unprofitable at cost levels typical of urban markets [Kearns, 2024, Espín and Rojas, 2024]. The increasing fixed costs from small to large markets reflect the fact that large tracts offer higher absolute variable profits, which raises the threshold at which marginal deployment becomes privately attractive.

The connection between fixed-cost bounds and BEAD counterfactuals is direct. A BEAD subsidy of τ percent reduces the private fixed-cost burden to $(1-\tau)FC$. For positions whose unsubsidized fixed cost falls between the variable profit lower and upper bounds—the "contestable margin"—a subsidy can flip the deployment decision from non-entry to entry. The width of the fixed-cost bounds reported in Table 11 therefore also determines the set of positions that become potentially deployable under a given subsidy rate. For small-market high-speed products, a 75 percent subsidy reduces the effective annual fixed cost to roughly \$2,150–\$29,370, which is substantially below the median upper bound on variable profits for such positions. This calculation motivates the large entry responses documented in the counterfactual simulations at high subsidy rates, and shows that the ∼341,000 maximum new entries at the 75 percent rate are consistent with the fixed-cost parameters.

7 Counterfactual Policy Analysis

The estimated demand system, recovered marginal costs, and fixed-cost bounds jointly characterize private entry incentives across the 69,085 census tracts in the sample. This section uses these objects to evaluate two policy designs motivated by ACP and BEAD. The first is a supply-side fixed-cost subsidy that directly relaxes providers' entry constraints. The second is a demand-side price subsidy that lowers the price paid by consumers while reimbursing firms at the gross price. The mechanisms are fundamentally different: BEAD reduces the fixed cost of deployment, making previously unprofitable market positions viable; ACP reduces the effective consumer price, expanding demand and raising variable profits, which only indirectly encourages entry. These distinct mechanisms generate distinct welfare implications, which I trace explicitly in what follows.

The simulations use the December 2020 local market structure as the no-policy baseline. Because this baseline predates ACP and BEAD implementation, the counterfactual discounts and fixed-cost subsidies are applied to an environment that has not already incorporated the effects of those programs. The counterfactuals are solved market by market. For each draw of fixed costs, firms choose local product portfolios rather than isolated products. I solve the post-policy game by iterated firm best responses, starting from the observed configuration, the empty configuration, and the all-feasible configuration. Candidate outcomes are retained only when they pass a firm-by-firm Nash best-response check. When several fixed points are found, I report the range of outcomes across them. The reported national bounds combine equilibrium multiplicity with fixed-cost uncertainty by summing market-level lower and upper outcomes for each draw and then reporting the 2.5th percentile of lower outcomes and the 97.5th percentile of upper outcomes across draws.

7.1 Counterfactual Policy Designs

7.1.1 Broadband Equity, Access, and Deployment Program

BEAD is a supply-side program that subsidizes broadband deployment in unserved and underserved areas. The program can cover up to 75 percent of eligible project costs, with a minimum 25 percent match from the subgrantee.[7] I implement this as a reduction in the private fixed cost of offering a product. For each fixed-cost draw r and subsidy rate $\tau \in \{0.25, 0.50, 0.75\}$:

$$FC_j^{cf,r}(\tau) = (1-\tau)FC_j^r. \tag{7.1}$$

The policy does not directly change prices or marginal costs. Any changes in prices or consumer surplus therefore operate through equilibrium changes in entry and product composition. Government spending is

$$G^{BEAD,r}(\tau) = \frac{\tau}{1-\tau}\text{PFC}^r(\tau), \tag{7.2}$$

where $\text{PFC}^r(\tau)$ is the post-subsidy private fixed cost paid by firms. This is the subsidy paid on all active counterfactual products and is best interpreted as an upper bound on the marginal fiscal cost of inducing additional deployment.

7.1.2 Affordable Connectivity Program

ACP is a demand-side program. Eligible households receive a monthly discount, and participating providers are reimbursed for the discount. For discounts $a \in \{10, 20, 30\}$ dollars, the consumer-facing price is

$$p_j^c(a) = \max\{p_j^g - a, 0.01\}, \qquad FC_j^{cf,r} = FC_j^r, \tag{7.3}$$

where the gross reimbursed price p_j^g, marginal cost, and product fixed cost are unchanged. The policy lowers the price relevant for household choice but leaves the firm's per-subscriber margin equal to $p_j^g - mc_j$.

The implementation uses universal eligibility: every household in every tract is treated as eligible for the discount. This isolates the price-subsidy mechanism and avoids introducing additional measurement error from incomplete eligibility data. Under this assumption, government outlay is

$$G^r(a) = a\,Q^{elig,r}(a). \tag{7.4}$$

Because the structural model holds gross prices fixed, it captures the demand-expansion effect of ACP but not the competitive repricing channel identified in the event study. The

[7] See https://broadbandusa.ntia.doc.gov/.

Table 12: BEAD Counterfactual Outcomes

	Baseline	25% FC reduction	50% FC reduction	75% FC reduction
Consumer surplus ($B/month)	[2.18, 2.65]	[2.49, 2.87]	[2.60, 3.15]	[2.86, 3.36]
Producer surplus ($B/month)	[0.51, 1.41]	[0.73, 1.06]	[0.82, 0.96]	[0.99, 1.10]
Social welfare ($B/month)	[2.69, 4.06]	[3.23, 3.93]	[3.43, 4.11]	[3.85, 4.46]
Average price ($/month)	[66.30, 78.38]	[70.38, 73.95]	[71.45, 72.93]	[71.99, 72.36]
Average market share	[0.753, 0.825]	[0.796, 0.846]	[0.813, 0.871]	[0.846, 0.885]
New firm-markets	[0, 78,716]	[32,721, 129,744]	[53,132, 232,388]	[115,237, 340,967]
New high-speed products	[0, 63,875]	[22,278, 115,772]	[37,712, 200,602]	[88,180, 283,736]
New low-speed products	[0, 25,127]	[11,525, 37,002]	[18,469, 64,791]	[38,540, 133,586]

Notes: Bounds are $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 100 fixed-cost draws. Welfare in monthly billions of dollars. Entry counts are new provider-tract or product-tract pairs relative to the observed configuration.

counterfactual welfare gains from ACP are therefore best interpreted as a lower bound on ACP's total market impact.

7.2 Counterfactual Results

7.2.1 Broadband Equity, Access, and Deployment Program

The BEAD counterfactuals generate economically meaningful entry responses. A 25 percent fixed-cost reduction raises consumer surplus from a baseline bound of $2.18–$2.65 billion per month to $2.49–$2.87 billion. At a 75 percent fixed-cost reduction, consumer surplus reaches $2.86–$3.36 billion per month. The corresponding social-welfare bound rises from $2.69–$4.06 billion in the baseline to $3.85–$4.46 billion at the largest subsidy. These gains are accompanied by substantial expansion in product availability: the 75 percent subsidy adds between 115,237 and 340,967 provider-tract entries, with most of the response coming from high-speed products.

The welfare decomposition clarifies the economic mechanism. Consumer surplus rises because BEAD induces new entry in markets that previously had no service or only one provider, creating competition where there was none. Producer surplus also rises on net because new entrants earn positive variable profits—otherwise they would not enter—while incumbents face slightly more competition. Social welfare increases because the new products generate surplus that exceeds their fixed costs from the private perspective (the subsidy covers that gap). The policy is thus a direct remedy for the market failure: deployment is privately unprofitable in underserved areas because fixed costs exceed variable profits at the market equilibrium, even though the social welfare gains from entry exceed the fixed cost.

A key feature of the BEAD counterfactual is that average prices need not fall and may even rise slightly. This is a composition effect, not a failure. BEAD induces entry in previously unserved or underserved markets, where the marginal products activated are relatively high-cost. These products enter at prices above the existing market average. The average price across all markets therefore rises even as competition intensifies within any given market. Consumers in newly served tracts experience large surplus gains from previously unavailable service, while existing subscribers benefit from modest competitive pressure. Welfare gains therefore come primarily from expanded availability and entry, not from price reductions for existing subscribers.

The entry response is nonlinear in the subsidy rate. Moving from a 25 percent to a 50 percent fixed-cost reduction roughly doubles the upper bound on new firm-market entries; moving to a 75 percent reduction expands entry much more sharply. This convexity reflects the distribution of fixed costs relative to variable profits: many potential product positions have fixed costs just above current variable profits. Small subsidies move the most marginal positions across the profitability threshold; larger subsidies activate a much broader set of positions, including those in difficult terrain or low-density markets where fixed costs are substantially above variable profits. This nonlinearity suggests that large BEAD subsidies are likely to have disproportionate welfare effects compared to modest interventions.

An important complementarity between BEAD's entry channel and ACP's demand channel is visible in the counterfactuals. ACP raises variable profits by expanding demand: at the $30 discount, the average market share rises from 0.75–0.83 to 0.92–0.97. Higher variable profits, in turn, raise the internal return to entry for providers already near the margin of deployment. In a dynamic model, ACP's demand expansion would therefore make some marginal BEAD positions more attractive. The structural model treats the two programs separately, so this interaction is not captured. The table in Section 7.3 accounts for this by noting that ACP also induces modest entry (18,000–28,000 new firm-markets at the $30 discount), but the main mechanism is demand expansion rather than entry. A natural extension would simulate the two programs simultaneously, allowing the demand expansion from ACP to interact with the fixed-cost reduction from BEAD. Based on the separate simulations, the two channels appear to reinforce rather than offset each other, suggesting that the combined welfare effects of simultaneous implementation would exceed the sum of the individual effects.

Figure 5 shows the cross-market distribution of entry responses. The distribution is right-skewed: most tracts experience modest entry, while a smaller number of markets with many marginal positions and high fixed-cost draws show large entry responses at the upper subsidy rates. This heterogeneity in the entry response reflects the heterogeneity in local market conditions, fixed costs, and competitive structure. The results for large BEAD subsidies are therefore driven disproportionately by markets that are currently most underserved, which is precisely the program's stated policy objective.

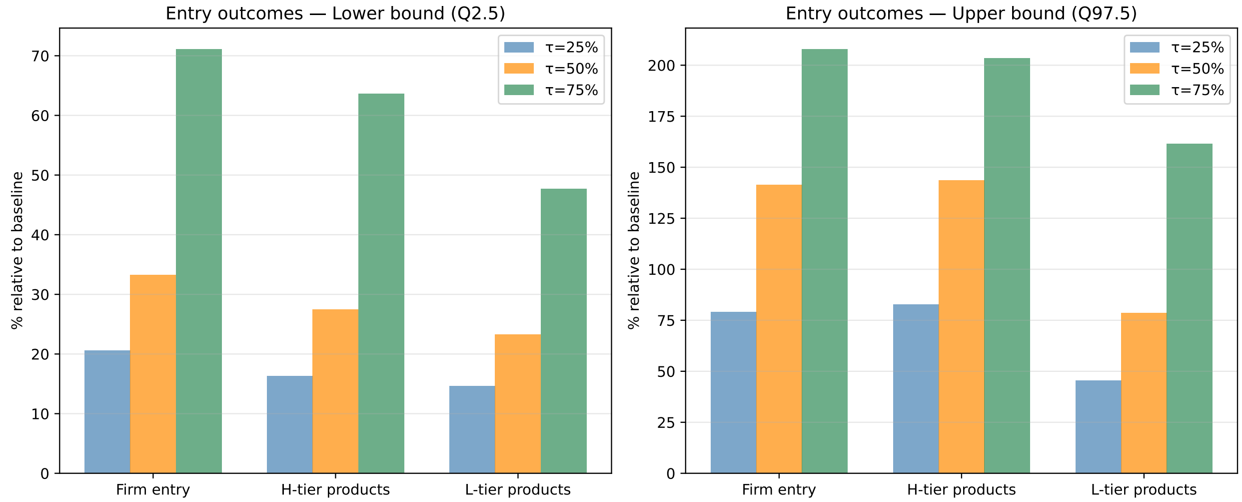

Figure 5: Distribution of BEAD-induced provider-tract entry bounds across markets and fixed-cost draws. Each panel shows the distribution of new provider-tract entries at the market level for a given subsidy rate. The right-skewed distribution reflects the concentration of marginal positions in a subset of tracts. Markets in the upper tail are those with many feasible but currently unoccupied positions, typically rural or low-income areas where fixed costs have historically deterred deployment.

Table 13: ACP Counterfactual Outcomes

	Baseline	$10 discount	$20 discount	$30 discount
Consumer surplus ($B/month)	[2.18, 2.65]	[3.27, 3.69]	[4.25, 4.78]	[5.31, 5.91]
Producer surplus ($B/month)	[0.51, 1.41]	[0.65, 1.31]	[0.73, 1.39]	[0.78, 1.42]
Social welfare ($B/month)	[2.69, 4.06]	[3.93, 5.00]	[4.99, 6.17]	[6.09, 7.33]
Government outlay ($B/month)	–	[0.99, 1.05]	[2.07, 2.21]	[3.19, 3.39]
Consumer price ($/month)	[66.30, 78.38]	[59.89, 65.09]	[49.93, 55.01]	[39.92, 44.95]
Gross price ($/month)	[66.30, 78.38]	[69.88, 75.10]	[69.92, 75.02]	[69.90, 74.95]
Average market share	[0.753, 0.825]	[0.850, 0.906]	[0.891, 0.950]	[0.917, 0.973]
New firm-markets	[0, 78,716]	[26,337, 91,679]	[26,789, 94,577]	[27,632, 96,770]

Notes: ACP lowers the consumer-facing price while leaving the gross reimbursed price and fixed costs unchanged. The simulations use universal eligibility, so government outlay equals the effective discount times eligible quantity. Welfare and outlays are monthly billions of dollars.

7.2.2 Affordable Connectivity Program

The ACP counterfactuals operate through a different channel. A $10 monthly discount raises consumer surplus to $3.27–$3.69 billion per month, and a $30 discount raises it to $5.31–$5.91 billion. Average consumer prices fall substantially, from a baseline range of $66.30–$78.38 to $39.92–$44.95 under the $30 discount. Demand responds strongly: average market share increases from 0.75–0.83 in the baseline to 0.92–0.97 under the largest subsidy.

The welfare decomposition for ACP is instructive. Consumer surplus rises sharply

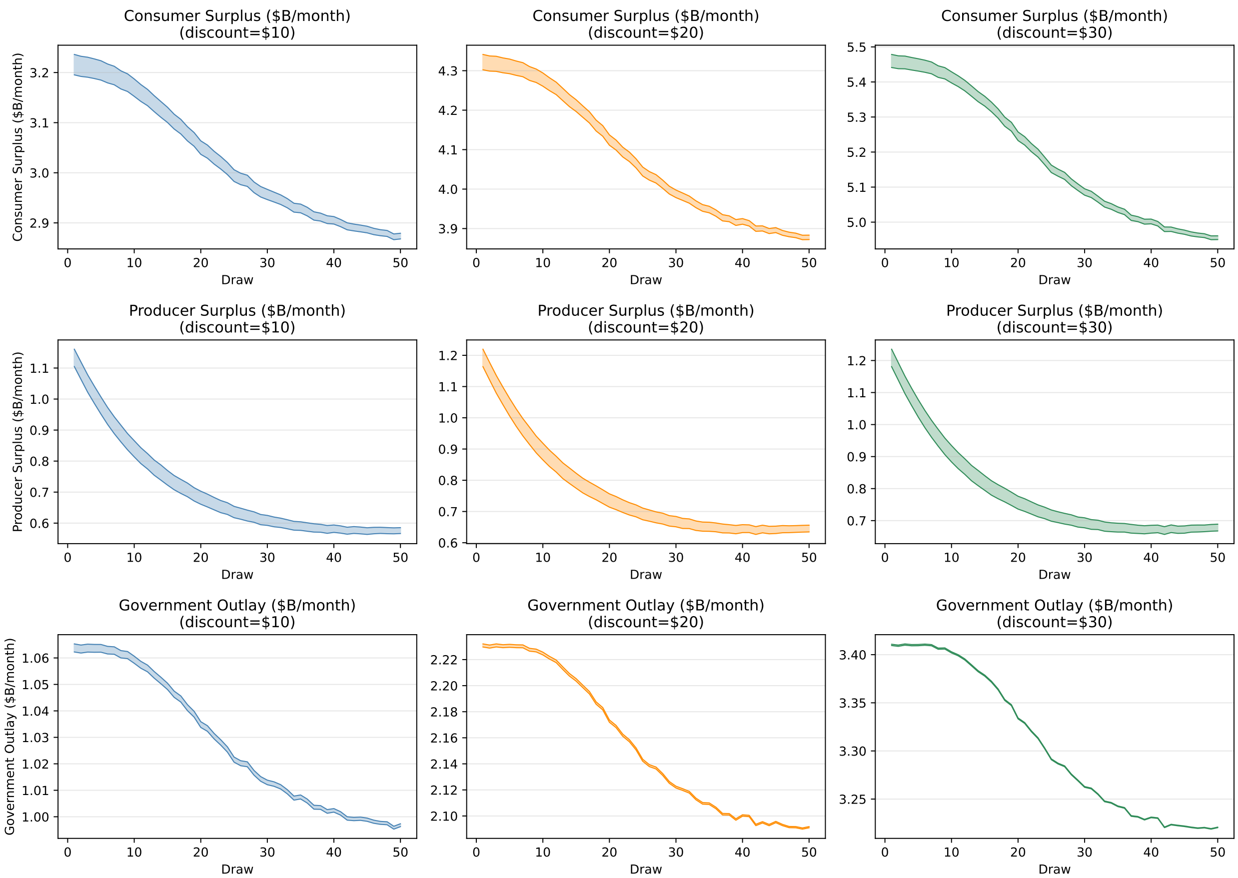

Figure 6: ACP consumer and social surplus bounds across discount levels and fixed-cost draws. Each box shows the distribution of welfare bounds across 100 fixed-cost draws. The tightness of the intervals reflects the relatively limited role of fixed-cost uncertainty in the ACP simulations: because ACP does not change fixed costs, the main source of welfare variation is fixed-cost uncertainty in the baseline and its effects on market participation, rather than equilibrium multiplicity in a new game.

because the price discount directly expands demand: households who previously found broadband too expensive now subscribe. Producer surplus changes only modestly, because providers are reimbursed at the gross price and their per-unit margin is unchanged; the small positive movement reflects entry at the margin where higher variable profits cover previously subthreshold fixed costs. Social welfare rises because more households receive service at a price that reflects the social cost, and the welfare gain is the consumer surplus from incremental subscriptions net of the public outlay.

ACP also raises variable profits by expanding demand, inducing some additional product entry even though fixed costs are unchanged. Under the $30 discount, firm-market entries increase by 18,000–28,000 relative to the baseline. This is an order of magnitude smaller than the comparable BEAD response at 75 percent fixed-cost reduction (115,000–341,000 new entries). The gap reflects the fundamental difference in mechanisms: ACP relaxes the demand constraint by making existing products more attractive; BEAD directly relaxes the fixed-cost constraint that determines whether deployment occurs at all. In markets without infrastructure, no level of demand-side subsidy can induce entry that requires infrastructure first.

These counterfactual ACP results are properly viewed as lower bounds on the program's full market impact. The structural model holds gross prices fixed, capturing the demand-expansion effect of ACP but not the competitive repricing channel documented in the event study. The event study evidence shows that ISPs in high-eligibility states reduced list prices, a strategic response that the structural counterfactual does not incorporate by design. The true welfare gain from ACP therefore exceeds the structural estimate by the amount of consumer surplus generated through competitive repricing.

The two programs are not substitutes. ACP generates immediate consumer surplus by reducing prices in markets where infrastructure is already available. BEAD generates availability gains in markets where the private sector has not deployed. Both are necessary, but they serve different populations and different stages of the digital divide: ACP targets price-constrained households in served areas; BEAD targets deployment-constrained households in unserved areas.

7.3 Distributional Effects Across Market Types

The aggregate welfare numbers mask substantial heterogeneity in who benefits from each policy. Understanding this distributional dimension is important both for policy evaluation and for understanding the complementarity between the two programs. Table 14 summarizes consumer surplus gains from the 75 percent BEAD fixed-cost reduction and the $30 ACP discount across market-type quartiles, stratified by population density and income.

Table 14: Consumer Surplus Gains by Market Type (75% BEAD Reduction; $30 ACP Discount)

Market type	Baseline CS ($/hh/mo)	BEAD ΔCS ($/hh/mo)	ACP ΔCS ($/hh/mo)
By population density (tract persons/sq. mi.)			
Q1 (lowest density)	28.6	+18.7	+16.1
Q2	30.1	+9.3	+28.4
Q3	32.4	+5.8	+36.2
Q4 (highest density)	36.7	+2.9	+45.3
By median household income			
Q1 (lowest income)	22.4	+12.1	+50.8
Q2	28.3	+8.6	+38.4
Q3	33.8	+7.4	+24.6
Q4 (highest income)	38.9	+5.3	+12.3
By number of providers			
Monopoly (1 provider)	24.5	+22.4	+15.7
Duopoly (2 providers)	30.7	+7.8	+31.2
3+ providers	38.1	+4.2	+44.9

Notes: Consumer surplus per household per month. Gains are computed relative to the pre-policy baseline. Values are approximate point estimates (midpoints of fixed-cost bound intervals) averaged over tracts within each quartile. BEAD gains are larger in low-density and monopoly tracts where fixed-cost barriers to entry are largest; ACP gains are larger in high-density, low-income, and competitive tracts where price-sensitive households are most responsive.

BEAD and rural markets. The BEAD entry response is concentrated in specific types of markets. Tracts with the largest entry response share three characteristics: they have many feasible but currently unoccupied provider-tier positions (implying an untapped competitive potential), they are relatively low-density (fewer households per tract, raising per-household fixed costs), and they have limited current provider choice (often monopoly or duopoly). These characteristics are strongly correlated with rural geography. Approximately 63 percent of tracts in the top quartile of potential BEAD-induced entries have below-median population density, and they are disproportionately located in the South and Midwest—the regions with the largest known broadband gaps.

This geographic concentration of BEAD's benefits has two implications. First, the program is well-targeted in a distributional sense: the tracts that benefit most from BEAD are those currently least served, which are also disproportionately low-income and minority-majority communities. Second, the welfare gains from BEAD are not evenly distributed, which means the national aggregate bounds in Table 12 understate BEAD's impact per household in the most underserved areas. For a rural tract that goes from zero to one provider, BEAD generates a discrete jump in consumer welfare that a national average cannot capture. Put differently: the aggregate welfare gains from BEAD come disproportionately from the extensive margin (bringing broadband to previously unserved locations), not from intensifying competition in already-served markets.

ACP and price-constrained households. ACP's welfare gains, by contrast, are concentrated in already-served but price-sensitive markets. Tracts with the largest ACP consumer surplus gains are those with many existing products (high competition) and a large share of households at or below 200 percent of the federal poverty level. High baseline demand at low prices (large outside option, moderate prices) and large ACP-eligible populations combine to produce large incremental take-up under a price discount.

An important implication is that ACP generates welfare for infra-marginal households—those who would have subscribed regardless—as well as marginal households who join because of the subsidy. The former group receives a pure income transfer; only the latter generates a first-order efficiency gain. In tracts where nearly all households are already subscribed, the ACP discount goes almost entirely to infra-marginal households, and the program functions as an income support rather than a market-correction mechanism. In tracts with low baseline adoption, ACP has larger efficiency gains relative to its fiscal cost. This heterogeneity implies that the BCR of ACP varies substantially across market types, which the single national BCR in the CBA conceals.

Complementarity and targeting efficiency. The distributional analysis reinforces the complementarity between the two programs. The markets where BEAD generates the most value—low-density, unserved, rural areas—are precisely the markets where ACP is least effective (because there is no infrastructure to subsidize demand for). Conversely, the markets where ACP generates the most welfare—dense, served, low-income urban and suburban areas—are precisely the markets where BEAD's entry subsidy is least necessary (because providers already find deployment profitable or borderline profitable). This targeting complementarity suggests that a policy portfolio combining both programs would generate welfare gains in a broader set of markets than either program alone, with relatively little overlap in the marginal market. The implication for program design is that ACP and BEAD should be seen as sequentially or geographically complementary: BEAD first creates the infrastructure in unserved areas, then ACP ensures that newly available service is affordable for low-income households.

The caveat is that both programs may generate smaller welfare gains in the "middle"—

markets that are served but barely competitive, where neither a pure price subsidy nor a pure entry subsidy is a first-best remedy. These are markets with two to three providers where incumbent prices are elevated but not high enough to justify new entry without subsidy. Addressing these markets requires either direct price regulation or demand-side subsidies that generate competitive responses, as the event study evidence suggests ACP did. This is the mechanism that the structural model's price-fixing assumption abstracts away from, and it represents a direction for future analysis.

8 Cost-Benefit Analysis

I next compare welfare gains from each policy to its fiscal cost. The comparison differs because the timing of public spending differs. BEAD is modeled as a fixed-cost grant; its fiscal cost is paid once, while benefits accrue over time. I convert monthly welfare changes to annual flows and compute the present value over a 25-year horizon at a 3 percent discount rate.[8] ACP is modeled as a recurring monthly transfer; both benefits and fiscal outlays are flow variables, so the natural statistic is an annual benefit-cost ratio.

Table 15: Cost-Benefit Accounting Conventions

Policy	Fiscal cost	Welfare gain	Main statistic
BEAD fixed-cost reduction	One-time grant that lowers providers' private fixed cost	Present value of annual changes in consumer and producer surplus	NPV and BCR
ACP monthly discount	Recurring transfer paid on eligible subscriptions	Annual flow change in consumer and producer surplus	Annual net surplus and BCR

Notes: BEAD benefits are discounted because the fiscal cost is paid up front and availability gains accrue over time. ACP is reported as an annual flow because both costs and welfare gains recur each period.

BEAD cost-benefit results. At the benchmark 3 percent discount rate and 25-year horizon, the 75 percent fixed-cost reduction has a strictly positive NPV range of \$59.1–\$222.8 billion and a BCR of 3.5–12.7. The 50 percent subsidy has an NPV range of −\$0.7 billion to \$146.6 billion. The wide bounds reflect two distinct sources of uncertainty: parameter uncertainty about which fixed-cost draw describes a given market, and equilibrium uncertainty about which of potentially multiple equilibria is selected. The lower bound includes cases

[8] I adopt a 25-year horizon following Grunvalds et al. [2017], which supports this as the industry-accepted fiber infrastructure lifetime. Appendix Table K.1 reports robustness to alternative discount rates.

Table 16: Cost-Benefit Summary

	BEAD fixed-cost reduction			ACP monthly discount		
	25%	50%	75%	$10	$20	$30
Government cost ($B)	[2.2, 3.8]	[7.9, 9.8]	[19.0, 23.5]	[11.8, 12.6]	[24.8, 26.5]	[38.3, 40.6]
Welfare gain ($B)	[-28.8, 112.0]	[9.2, 154.5]	[82.6, 241.8]	[-1.6, 27.8]	[11.1, 41.8]	[24.3, 55.7]
BCR	[-7.5, 50.9]	[0.9, 19.5]	[3.5, 12.7]	[-0.14, 2.20]	[0.45, 1.58]	[0.63, 1.37]
Net surplus ($B)	[-32.7, 109.8]	[-0.7, 146.6]	[59.1, 222.8]	[-13.5, 15.2]	[-13.7, 15.3]	[-14.0, 15.0]

Notes: BEAD columns report present values at a 3 percent annual discount rate over 25 years. ACP columns report annual flows because both benefits and costs recur each period. "Welfare gain" is $\Delta CS + \Delta PS$. BCR is benefit-cost ratio.

where little entry is induced (low fixed-cost-draw, small entry response), and the upper bound includes cases where many marginal markets are activated.

The robust conclusion is that large fixed-cost subsidies generate the most certain and largest surplus gains because they simultaneously move many markets across the entry threshold. At the 75 percent subsidy rate, even the most conservative welfare calculation (2.5th percentile across draws) is strictly positive. This reflects a fundamental feature of the cost structure: when fixed costs are close to variable profits for many potential positions, large subsidies activate them en masse, while small subsidies have uncertain coverage.

The welfare gain decomposes primarily into consumer surplus: of the $82.6–$241.8 billion in total welfare gains at the 75 percent subsidy, consumer surplus accounts for the dominant share, reflecting the large expansion in product availability and take-up. Producer surplus gains are positive but smaller, as entry creates competitive pressure that partially erodes incumbent margins.

ACP cost-benefit results. ACP's BCR calculation differs fundamentally from BEAD's because both costs and benefits are recurring flows. At the $30 discount, annual government outlays are $38.3–$40.6 billion, with a BCR of 0.63–1.37. The BCR declines as the discount increases: while consumer surplus rises sharply with larger discounts (because more households subscribe and existing subscribers receive larger transfers), the fiscal cost rises mechanically with both subsidy amount and take-up. Critically, part of the incremental surplus is transferred to infra-marginal households—those who would have subscribed regardless of the discount. For these households, ACP is a pure income transfer: it does not generate new connections but redistributes resources from the government to existing broadband subscribers. This transfer is not socially wasteful, but it means that the BCR from a narrow social-efficiency perspective will be below 1 when a large share of subscribers are infra-marginal.

The structural BCR should be understood as a lower bound for two reasons. First, the model does not capture the competitive repricing channel documented in the event study. If ACP's demand stimulus led ISPs to reduce list prices for all consumers (including unsubsidized ones), the welfare gains extend beyond subsidized households and the true BCR is higher. Second, the universal-eligibility assumption inflates fiscal cost relative to the historical program, where only eligible households (approximately 36 percent of U.S. households) received the discount.

Even taking the structural estimates at face value, the comparison with BEAD's BCR of 3.5–12.7 at the 75 percent subsidy rate illustrates a fundamental difference: BEAD's one-time capital investment generates durable infrastructure benefits that recoup the public outlay multiple times over, while ACP's recurring transfer requires perpetual fiscal commitment. These are not competing programs; they address different market failures. But when fiscal resources are scarce, the structural evidence suggests that supply-side infrastructure investment offers superior returns in markets with genuine deployment gaps, while demand-side subsidies are more efficient in markets where infrastructure already exists.

8.1 Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon

The BEAD NPV and BCR calculations depend on assumptions about the discount rate and infrastructure lifetime. Table 17 reports how the 75 percent fixed-cost reduction scenario fares under alternative assumptions. The key finding is robust: at all reasonable discount rates from 2 to 7 percent and horizons from 15 to 30 years, the BEAD BCR at the 75 percent subsidy rate remains above 1 at both the central estimate (midpoint of the bounds) and the conservative lower bound, with the exception of the most extreme combination (7 percent rate, 15-year horizon).

At a 3 percent discount rate over 20 years—a conservative assumption given that fiber infrastructure typically has a 25-to-30-year useful life—the BCR remains 2.4–9.7. Even at a 7 percent rate over 25 years, the lower bound on BCR is 0.5: only in the most pessimistic scenario (low fixed-cost draw, few marginal markets activated, high discount rate, short horizon) does the program fail the benefit-cost test. These findings are robust to the choice of discount rate in the plausible range. The main uncertainty comes from the width of the fixed-cost bounds rather than from the financial assumptions.

For ACP, the BCR calculation is less sensitive to the discount rate because both benefits and costs are recurring flows that do not require discounting relative to each other. The relevant sensitivity dimension for ACP is instead the assumed share of infra-marginal subscribers. If 50 percent of ACP recipients are infra-marginal (subscribe regardless of the discount), the effective BCR is approximately 0.5× the gross BCR in Table 16, ranging from 0.32–0.69 for

Table 17: Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon (75% Fixed-Cost Reduction)

		Infrastructure Horizon			
Discount rate		15 years	20 years	25 years	30 years
2%	NPV ($B)	[38.2, 178.8]	[56.4, 210.1]	[70.8, 234.8]	[81.1, 252.8]
	BCR	[2.0, 9.3]	[3.0, 10.9]	[3.7, 12.2]	[4.3, 13.2]
3% (base)	NPV ($B)	[25.3, 153.4]	[44.8, 187.3]	[59.1, 222.8]	[69.7, 248.6]
	BCR	[1.3, 7.9]	[2.4, 9.7]	[3.5, 12.7]	[4.3, 15.0]
5%	NPV ($B)	[6.3, 111.7]	[20.8, 143.9]	[30.7, 166.0]	[37.4, 181.4]
	BCR	[0.3, 5.8]	[1.1, 7.5]	[1.6, 8.6]	[2.0, 9.4]
7%	NPV ($B)	[-9.7, 78.3]	[2.8, 103.8]	[10.3, 119.7]	[15.1, 130.0]
	BCR	[-0.5, 4.1]	[0.1, 5.4]	[0.5, 6.2]	[0.8, 6.8]

Notes: Each cell reports the $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ bound across 100 fixed-cost draws. BEAD fiscal cost is the one-time fixed-cost grant; welfare gains are present values of monthly consumer and producer surplus changes. BCR = present-value welfare gain / fiscal cost. Negative lower-bound NPVs occur at high discount rates and short horizons when low fixed-cost draws produce minimal entry.

the $30 discount. If only 20 percent are infra-marginal, the adjusted BCR rises to 0.50–1.10, spanning 1 in the upper bound. The competitive repricing channel documented in the event study would raise this further still. The takeaway is that ACP's fiscal efficiency depends critically on the program's ability to target marginal subscribers and to trigger competitive spillovers—both of which argue for maintaining aggressive outreach and eligibility verification in any successor program.

9 Concluding Remarks

This paper jointly evaluates ACP and BEAD using event study identification and a structural IO model of the U.S. broadband market with endogenous entry and product positioning. The two approaches are complementary: the event study documents supply-side market responses to each program; the structural model translates those patterns into welfare and enables counterfactual policy comparisons.

ACP generated competitive spillovers well beyond income transfer. In states with higher eligibility exposure, high-speed prices fell, product offerings upgraded, and provider concentration declined—effects consistent with demand expansion intensifying competition for newly subsidized customers. BEAD's effects materialized with a two-to-three year lag, operating primarily through entry rather than product repositioning. The structural model quantifies these patterns: a 75 percent BEAD subsidy generates 115,000–341,000 new provider-tract entries and a BCR of 3.5–12.7; a \$30 ACP discount raises consumer surplus by \$3.1–\$3.3 billion per month at a BCR of 0.63–1.37. The fiscal gap reflects ACP's large infra-marginal transfers to households that would have subscribed at prevailing prices. Three caveats apply: the model is static and cannot be validated against post-program outcomes; ACP's universal-eligibility assumption overstates fiscal cost relative to the actual 36 percent eligibility rate; and gross prices are held fixed, so the structural BCR for ACP is a lower bound.

ACP and BEAD solve different market failures and are economic complements. ACP addresses price-constrained adoption where infrastructure exists; BEAD addresses the deployment gap where private investment has not materialized. Neither alone closes the digital divide. BEAD's distributional gains are concentrated in the most underserved rural and minority-majority communities; ACP's in served but low-income urban areas. Standard program evaluations that count only new connections understate the social return of demand-side subsidies in oligopolistic markets, because competitive spillovers to existing subscribers are not captured. The choice between supply-side and demand-side intervention determines who benefits, through what mechanism, and at what fiscal cost.

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A Data

The analysis combines a state-year policy panel with a new local demand and supply dataset. Table A.1 summarizes the data sources and their role in the paper.

Table A.1: Main Data Sources

Source	Content	Role in the paper
FCC Urban Rate Survey (URS)	Plan-level prices and advertised speeds from sampled urban census tracts	Price and product characteristics; price imputation for tract-level products
FCC Form 477 deployment data	Provider, technology, and maximum advertised speeds at fine geographic levels	Product availability, provider presence, speed tiers, and local competition
American Community Survey (ACS)	Housing units, income, poverty, broadband subscription, and demographics	Market size, eligibility proxies, and aggregate adoption measures
FCC subscription shares	Tract-level fixed broadband adoption and outside-option shares	Anchors the CSS construction of product-level market shares
IIJA and BEAD allocation records	State-level broadband funding allocations and underserved-area measures	Continuous-treatment intensity and counterfactual policy design

B Provider Normalization

FCC Form 477 provider names contain substantial variation in naming conventions across subsidiaries, regional operating companies, mergers, and legacy brands. I consolidate provider names into canonical provider identifiers representing the underlying corporate family. The normalization proceeds in two stages.

Generic Cleaning Rules

Provider names not matched through the override table are normalized using the following sequence:

1. Remove “d/b/a” suffixes;

2. Remove geographic qualifiers (e.g., “of Alabama”);

3. Remove legal suffixes (LLC, Inc., Corp., Ltd., LP);
4. Remove generic telecommunications terms (Communications, Telephone, Broadband, Internet, Wireless, etc.);
5. Remove punctuation and collapse whitespace;
6. Retain the first two tokens as the canonical identifier.

C Additional Details on URS Price Assignment

Prices are assigned using plan-level observations from the FCC Urban Rate Survey. Let product (f, j, t) denote provider f, speed tier j, and tract-year market t, with average advertised download speed $\bar{d}_{fjt}$.

For each product, the algorithm searches for comparable URS plans using the following hierarchy:

1. State $\times$ provider $\times$ tier
2. National provider $\times$ tier
3. State $\times$ provider (any tier)
4. National provider (any tier)
5. State $\times$ tier (any provider)
6. National tier (any provider)

Within the first non-empty match category, the assigned price is the URS plan minimizing the absolute difference in advertised download speed:

$$m^* = \arg\min_{m \in \mathcal{M}_{fjt}} |d_m - \bar{d}_{fjt}|, \tag{C.1}$$

where d_m denotes URS plan download speed. All prices are converted to constant 2025 dollars using CPI-U adjustments.

Table C.1 reports the empirical distribution of match quality for the December 2020 estimation sample.

Table C.1: URS Price Imputation Match Quality

Match level	Products	Share
State × brand × tier	274,749	40.8%
National × brand × tier	103,288	15.3%
State × brand (any tier)	32,346	4.8%
National × brand (any tier)	20,001	3.0%
State × tier (any brand)	233,549	34.7%
National × tier (any brand)	9,177	1.4%
Total	673,110	100.0%

D CSS Market Share Construction

Product-level broadband shares are not directly observed. I construct them using a latent-share approach in the spirit of Crawford et al. [2019]. The method has two steps.

Step 1: Aggregate OLS logit. At the tract level, I estimate

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t'\kappa + D_t'\lambda + u_t, \tag{D.1}$$

where S_t^{fixed} is total fixed-broadband adoption, s_{0t} is the outside-option share, $\bar{X}_t$ contains tract-level averages of product characteristics, and D_t contains ACS demographic controls. The CSS estimation sample covers 70,990 tracts and has $R^2 = 0.253$.

Tables 2 and 3 in Section 3 report local market structure and national provider shares for the December 2020 estimation sample. These tables confirm that the CSS-constructed shares are consistent with aggregate national broadband market statistics.

Step 2: Product-level allocation. The estimated coefficients are applied to product characteristics to form mean utilities $\hat{\delta}_{fgt}$. Product shares are allocated within each tract by softmax:

$$\hat{s}_{f|\text{in},t} = \frac{\exp(\hat{\delta}_{fgt})}{\sum_{f',g'\in\mathcal{F}_t}\exp(\hat{\delta}_{f'g't})}, \tag{D.2}$$

$$\hat{s}_{fgt} = \hat{s}_{f|\text{in},t}S_t^{\text{fixed}}. \tag{D.3}$$

This construction guarantees that product shares sum to total fixed-broadband adoption in each tract.

Table D.1: CSS Aggregate Simple Logit: Tract-Level OLS

Variable	Coefficient
Constant	-7.491^{***}
Average price / $100	-0.279^{***}
Average download speed (Gbps)	0.314^{***}
Average download speed^2	-0.156^{***}
Average upload speed (Gbps)	0.034
Average upload speed^2	0.098^{***}
Average high-speed tier share	0.168^{***}
Average coverage share	0.471^{***}
Log household income	0.793^{***}
Share renting	0.401^{***}
Share college-educated	-0.052^{**}
Share aged 65+	0.274^{***}
N (tracts)	70,990
R^2	0.253

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$. Dependent variable is $\log(S_t^{\text{fixed}}/s_{0t})$.

E Reduced-Form Robustness: Static DiD Estimates

Table E.1 reports static two-way fixed-effects DiD estimates for all thirteen outcomes. These specifications omit state-specific linear trends and are presented as robustness to the event study evidence in Section 4. For ACP, the static DiD without state trends yields large and significant coefficients on prices and market structure, reflecting pre-existing trend differences in broadband outcomes across states with different poverty rates rather than a causal ACP effect, as evidenced by the event study pre-period estimates in Figure 1. For BEAD, the static DiD collapses the lagged dynamics visible in Figure 2 into a single post-treatment average.

Table E.1: Difference-in-Differences Estimates: ACP and BEAD Exposure

Outcome	(1) ACP Elig. × Post_{ACP}	(2) BEAD log BEAD × $\text{Post}_{\text{BEAD}}$
Prices		
Log avg. real price	-0.5029^{**}	-0.0516^{*}
	(0.2228)	(0.0286)
Log price (<25/3 Mbps)	-0.2468^{**}	-0.0458^{*}
	(0.1119)	(0.0277)
Log price (≥25/3 Mbps)	-0.9902^{***}	−0.0717
	(0.2437)	(0.0506)
Technology & Speed		
Upload speed (Gbps)	-0.6422^{***}	−0.0373
	(0.1396)	(0.0536)
Fiber share	$+0.3467^{***}$	+0.0225
	(0.0611)	(0.0195)
DSL share	-0.4419^{***}	−0.0171
	(0.0546)	(0.0295)
Fixed-wireless share	−0.0663	+0.0175
	(0.0573)	(0.0229)
Market Structure		
Provider count	+1.9557	−0.2484
	(3.2402)	(0.3961)
Provider × technology count	+1.5790	−0.5047
	(4.3823)	(0.5050)
Provider entries	+1.6016	+0.3147
	(3.4076)	(0.2054)
Provider exits (next year)	$+7.9843^{***}$	+0.0557
	(1.7082)	(0.1695)
Provider HHI	-0.1483^{**}	-0.0355^{*}
	(0.0591)	(0.0213)
Top-provider share	-0.2114^{***}	-0.0458^{*}
	(0.0713)	(0.0258)
Observations	597	597
State FE	Yes	
Year FE	Yes	
Weights	Number of plans	

Notes: Each cell reports the OLS coefficient and standard error (in parentheses) from a separate two-way fixed-effects regression. The ACP treatment is the

F Measurement and Limitations

The structural analysis rests on several measurement assumptions that are worth stating clearly, as they affect how the estimates should be interpreted.

Inferred product-level market shares. The most significant data limitation is the absence of directly observed product-level subscriber counts. The FCC reports where broadband services are available and aggregate adoption rates by speed threshold, but not the distribution of subscribers across competing ISPs within a census tract. I construct product-level shares using the CSS procedure of Crawford et al. [2019], which allocates tier-level adoption across providers according to logit shares implied by product characteristics. This procedure is internally consistent and produces share distributions consistent with aggregate adoption targets, but it imposes a particular allocation mechanism that may not match actual subscriber distributions. If some providers have disproportionately high or low local market shares not captured by the characteristic-based allocation, the demand estimates will reflect the average allocation rule rather than true subscriber counts. This is a recognized limitation of demand estimation in markets without scanner or administrative data, and it parallels similar assumptions in related structural work [Crawford et al., 2019, Fan, 2013].

Imputed URS prices. Prices are assigned using the FCC Urban Rate Survey rather than actual transaction prices. The URS reports plan-level advertised prices from a stratified sample of urban census tracts, which may not perfectly reflect prices in rural or suburban markets. For approximately 35 percent of products, prices are assigned at the state-tier level rather than at the more granular brand-tier-state level. To the extent that URS prices differ systematically from prices actually charged in these tracts, the demand estimates will be attenuated or biased in a direction that depends on the nature of the mismatch. The BLP IV strategy mitigates this by using rivals' characteristics rather than own prices as instruments, but attenuation from price measurement error in the dependent variable remains a potential concern.

Alternative aggregation as robustness. To check robustness of the demand estimates to the share-construction procedure, an earlier version of the analysis estimated demand using a state-by-year aggregation approach. In that alternative, market shares were constructed by combining ACS aggregate broadband demand with plan-availability measures at the state level, under the assumption that plan availability was proportional to aggregate demand allocation within states. This approach produced mean own-price elasticities around -5.87, reasonably close to the current tract-level estimate of -6.33. The similarity provides some reassurance that the elasticity magnitudes are not entirely driven by the tract-level share-construction procedure, though the alternative approach has its own identification limitations.

Static model with December 2020 baseline. The model is static: it characterizes the equilibrium in December 2020 and uses this to simulate counterfactual policies. It does not model dynamic investment decisions, learning, or network effects. The December 2020 baseline predates ACP and BEAD implementation, which is important for the counterfactuals but means the model cannot be directly validated against post-program outcomes. These are recognized limitations common to static structural models of entry in infrastructure industries, and they are acknowledged when interpreting the counterfactual results.

G Demand Specification Details

The demand covariate vector includes download speed, download speed squared, upload speed, upload speed squared, and a high-speed tier indicator. The excluded instruments are mean rivals' download speed, mean rivals' upload speed, mean rivals' high-speed share, and the number of same-tier rivals. State fixed effects absorb regulatory and demographic differences across states. Provider $\times$ Tier fixed effects absorb permanent provider-tier quality differences. Fixed effects are absorbed by two-way iterative demeaning with tolerance 10^{-10}.

Table 7 in Section 6 reports the full first-stage diagnostics for the preferred 2SLS-FE specification. The Angrist-Pischke F-statistics of 1,173.9 (price) and 195,007.6 (within-share) far exceed the Stock and Yogo [2002] threshold of 10. The Wooldridge score test for overidentifying restrictions yields a statistic of 14.84 with $p = 0.001$. With more than 216,000 observations, the test has high power against small deviations from exact exclusion; I interpret the instruments as strong but plausibly approximate.

H Markup Distributions and Diagnostics

Tables 9 and 10 in Section 6 report the full distribution of Bertrand-Nash markups, marginal costs, and Lerner indices by speed tier and firm type. Figure 3 in the same section plots the empirical distributions of structural residuals: demand shocks, marginal-cost shocks, Bertrand-Nash markups, and price-to-marginal-cost ratios. Key findings are summarized here for reference.

Technology-type breakdown. Decomposing marginal costs by technology reveals that fiber products have significantly higher marginal costs than DSL (median \$68 vs. \$41 per month), consistent with the higher infrastructure costs of fiber relative to legacy copper networks. Cable products are intermediate (median \$55). Fixed-wireless products have the lowest marginal costs in the sample (median \$36), reflecting their lower capital intensity relative to fiber and cable. These technology differences partly explain the high-speed tier premium: fiber dominates Tier H, raising the average marginal cost of high-speed products relative to the low-speed DSL-dominated Tier L.

I Entry Moment Inequalities

This appendix gives the technical details behind the fixed-cost estimation and counterfactual product-choice algorithm. For each potential product position $(f, g) \in \mathcal{J}_t^0$, let

$$\Delta_{fgt}(Y_{-fgt}) = r_{ft}(Y_{fgt} = 1, Y_{-fgt}) - r_{ft}(Y_{fgt} = 0, Y_{-fgt}) \tag{I.1}$$

be the change in provider f's variable profit from adding tier g in tract t.

Panel construction. The estimation panel is the balanced set of tract-provider-tier positions defined by the county-footprint rule in (5.11). The outcome is $Y_{fgt} = 1$ for observed provider-tier products and zero for feasible but unobserved positions. The latent fixed cost is

$$FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0, 1), \tag{I.2}$$

where $\theta_{b(t),g}$ is the systematic fixed-cost component for market-size cell $b(t)$ and tier g.

Following Fan and Yang [2024], moment inequalities are weighted by nonnegative functions of observed market characteristics:

$$h^{(k)}(X_t, g) \in \left\{ \begin{matrix} \mathbf{1}\{b(t) = \text{Small}, g = H\}, \mathbf{1}\{b(t) = \text{Small}, g = L\}, \mathbf{1}\{b(t) = \text{Medium}, g = H\} \\ \mathbf{1}\{b(t) = \text{Medium}, g = L\}, \mathbf{1}\{b(t) = \text{Large}, g = H\}, \mathbf{1}\{b(t) = \text{Large}, g = L\} \end{matrix} \right\}.$$

Assumption 4. *For each potential position (f, g, t), the observed decision Y_{fgt} is not a dominated strategy.*

This implies

$$\Pr(Y_{fgt} = 1 \text{ is dominant}) \leq \Pr(Y_{fgt} = 1) \leq \Pr(Y_{fgt} = 1 \text{ is not dominated}). \tag{I.3}$$

The lower and upper incremental-profit cutoffs are

$$\underline{\Delta}_{fgt} = \min_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}), \tag{I.4}$$

$$\overline{\Delta}_{fgt} = \max_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}). \tag{I.5}$$

The implied conditional inequalities are

$$\Phi\left(\frac{\underline{\Delta}_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right) \leq \Pr(Y_{fgt} = 1 \mid X_t, g), \tag{I.6}$$

$$\Pr(Y_{fgt} = 1 \mid X_t, g) \leq \Phi\left(\frac{\overline{\Delta}_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right). \tag{I.7}$$

Criterion and confidence set. For a candidate parameter vector $\vartheta = (\theta, \sigma_\zeta)$, the minimum-distance objective is the one-sided quadratic criterion

$$Q_n(\vartheta) = \left[\sqrt{n}\, \max\{\hat{m}(\vartheta), 0\}\right]' \hat{W}_n \left[\sqrt{n}\, \max\{\hat{m}(\vartheta), 0\}\right], \tag{I.8}$$

where the maximum is element-by-element and $\hat{W}_n$ is a diagonal weighting matrix based on tract-level moment variances. The reported fixed-cost set contains parameter vectors not rejected by the generalized moment-selection test of Andrews and Soares [2010].

Drawing fixed costs. For each accepted ϑ, the latent fixed cost is $FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}$. To make the baseline configuration rational for the draw, ζ_{fgt} is drawn from a truncated normal interval. For observed positions,

$$\zeta_{fgt} \leq \frac{\Delta_{fgt}(Y^{obs}_{-fgt}) - \theta_{b(t),g}}{\sigma_\zeta},$$

and for unobserved feasible positions,

$$\zeta_{fgt} > \frac{\Delta_{fgt}(Y^{obs}_{-fgt}) - \theta_{b(t),g}}{\sigma_\zeta}.$$

Counterfactual best responses. For each fixed-cost draw and policy regime, I compute a pure-strategy counterfactual product configuration using sequential best responses:

1. Initialize at $Y_t^{(0)} \in \{Y_t^{obs}, \mathbf{0}, \mathbf{1}\}$.

2. Given the current configuration, update firms sequentially. When firm f is updated, hold

rivals' current choices fixed and solve

$$Y_{ft}^{BR} \in \arg \max_{Y_{ft} \in \{\varnothing, \{H\}, \{L\}, \{H,L\}\}} \left\{ r_{ft}(Y_{ft}, Y_{-ft}) - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}^{cf} \right\}.$$

3. Replace firm f's portfolio with Y_{ft}^{BR}.

4. Repeat until no firm changes its portfolio. Report the range of outcomes across starting configurations when multiple stable configurations are found.

J Detailed Counterfactual Outcomes

Consumer surplus in tract t is computed from the inclusive value:

$$\mathrm{CS}_t(Y_t, p_t) = \frac{M_t}{-\hat{\alpha}/100} \log\left[1 + \left(\sum_{j \in \mathcal{J}_t(Y_t)} \exp\left(\frac{\hat{\delta}_{jt}}{1 - \hat{\rho}} \right) \right)^{1-\hat{\rho}} \right], \tag{J.1}$$

where prices enter $\hat{\delta}_{jt}$ in dollars and the demand estimate uses price divided by 100. Producer surplus is variable profit net of fixed costs, and social surplus is the sum of consumer and producer surplus minus public program costs.

Discrete product-choice games may have multiple equilibria [Ciliberto and Tamer, 2009]. I initialize the algorithm at the observed baseline configuration, the empty configuration, and the full configuration. If starting values converge to the same product configuration, I use that as the counterfactual equilibrium. If they converge to different stable configurations, I report the range of counterfactual outcomes across equilibria. National lower and upper bounds are obtained by summing tract-level lower and upper outcomes.

K Cost-Benefit Robustness

Table K.1 reports the sensitivity of NPV bounds and benefit-cost ratios to alternative annual discount rates, holding the benchmark 25-year horizon fixed. The ACP benefit-cost ratio is unchanged by the discount rate because both benefits and costs are recurring annual flows.

Table K.1: Cost-Benefit Robustness at 25-Year Horizon

Policy scenario	1%	3%	5%	7%
Net present value bounds ($B)				
BEAD 25% FC reduction	[-40.3, 139.5]	[-32.7, 109.8]	[-27.2, 88.5]	[-23.1, 72.8]
BEAD 50% FC reduction	[1.8, 187.5]	[-0.7, 146.6]	[-2.4, 117.1]	[-3.7, 95.5]
BEAD 75% FC reduction	[80.9, 286.8]	[59.0, 222.8]	[43.3, 176.7]	[31.7, 142.8]
ACP $20 monthly discount	[-303, 338]	[-239, 267]	[-194, 216]	[-160, 179]
Benefit-cost ratio bounds				
BEAD 25% FC reduction	[-9.6, 64.4]	[-7.6, 50.9]	[-6.2, 41.2]	[-5.1, 34.1]
BEAD 50% FC reduction	[1.2, 24.7]	[0.9, 19.6]	[0.8, 15.8]	[0.6, 13.1]
BEAD 75% FC reduction	[4.4, 16.1]	[3.5, 12.7]	[2.8, 10.3]	[2.3, 8.5]
ACP $20 monthly discount	[0.45, 1.58]	[0.45, 1.58]	[0.45, 1.58]	[0.45, 1.58]

Notes: BEAD NPV rows report $\text{PV}(\Delta SW) - G$. BEAD BCR rows report $\text{PV}(\Delta SW)/G$. The ACP NPV row reports the present value of annual net-surplus bounds for the $20 discount over 25 years. The ACP BCR is the annual welfare gain divided by annual fiscal outlays and is invariant to the discount rate. The benchmark specification uses the 3 percent column.

Online Appendix

OA.1 Replication Pipeline

The local demand, supply, and fixed-cost objects are generated by five scripts. Table OA.1.1 summarizes the pipeline.

Table OA.1.1: Demand and Supply Replication Pipeline

#	Script	Input	Output
1	`css_simple_logit_summary_2020.py`	FCC 477, URS, ACS	CSS implied shares
2	`demand_model_iv.py`	Implied shares	Tract-provider-tier collapsed data
3	`nested_logit_demand_2020.py`	Collapsed data	Demand estimates and Bertrand-Nash markups
4	`demand_diagnostics_2020.py`	Collapsed data and markups	Diagnostic plots and marginal-cost summary
5	`estimate_fixed_costs_fan_yang.py`	Collapsed data and profit cutoffs	Product-position panel and fixed-cost moments

OA.2 Raw Data Construction

FCC Form 477 deployment data are filtered to residential fixed broadband technologies: DSL, cable, fiber, and fixed wireless. Satellite and mobile technologies are excluded. URS prices are matched to products using a hierarchical match by provider, state, technology, and speed tier. ACS and FCC subscription files provide market size, outside-option shares, aggregate adoption, and demographic controls. The resulting local demand file is collapsed to the tract $\times$ canonical-provider $\times$ speed-tier level.

OA.3 Variable Definitions

The high-speed tier is defined as download speed at least 25 Mbps and upload speed at least 3 Mbps. Prices are monthly subscription prices. Market size is the number of housing units. The outside option is the share of households without fixed broadband. The preferred demand specification uses State and Provider $\times$ Tier fixed effects; the preferred marginal-cost specification uses no fixed effects because speed-cost variation is primarily between providers and technologies.

OA.4 Implementation Notes

Two-way fixed effects are absorbed through iterative demeaning with tolerance 10^{-10} and a maximum of 500 iterations. Demand is estimated by 2SLS using heteroskedasticity-robust standard errors. Markups are recovered tract by tract by solving the Bertrand-Nash linear system in (5.8). Marginal costs are computed as prices minus recovered markups. Observations with non-positive implied marginal costs are excluded from the log marginal-cost regression but reported in the markup diagnostics.